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Hybrid deep learning model for network intrusion detection using optimal feature fusion

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ABSTRACT

Recent Intrusion Detection (ID) networks face difficulties in handling the enlarging volume of network traffic and adapting to emerging cyber threats. Handling data traffic and addressing data imbalance are key requirements for identifying recent cyber threats. This paper proposes a novel hybrid ID system designed to mitigate data imbalance issues. The proposed methodology uses advanced deep learning techniques and optimized characteristic fusion models, making it suitable for high-traffic environments. This research conducts a comprehensive experimental study on five standard ID datasets, focusing on network traffic and system behavior data, which are crucial for detecting potential intrusions. In the deep feature extraction phase, multiple features are considered, including statistical information, T-SNE features, and high-level deep learning features. T-SNE features capture similarities between data points, helping preserve the most important features. For feature fusion, optimal weights are identified using the proposed RCMPA. The fused feature set, created from these optimized weights, improves that a more appropriate and discriminative characteristics are utilized for training. A system then employs a "Multi-scale Dilated Deep Hybrid Network with Attention Mechanism" (MDDHN-AM) for intrusion diagnosis. This developed model integrates TCNN and RNN to detain both temporal and spatial dependencies. TCNN processes sequential information to identify temporal patterns, while RNN captures the dynamic nature of network traffic. The attention mechanism prioritizes the most significant features, enabling more accurate intrusion detection. At last, the presentation of MDDHN-AM was compared to traditional and state-of-the-art intrusion detection methods across multiple metrics. The developed model achieved 96.43% detection accuracy and 97.38% precision, illustrating its efficiency in handling diverse digital attacks and data imbalance. An improved performance over traditional methods highlights its potential as a robust solution for secure communication and protection against evolving digital attacks.

1. Introduction

Network Intrusion Detection (NID) has been actively researched over the past decade and has shown potential improvements in defending against cyber threats. However, recent advances in digitalization have introduced several challenges, such as high data dimensionality, data imbalance, adversarial attacks, and the need for early detection [1]. Most recent studies in NID have focused on addressing data imbalance by increasing intrusion traffic, since, in typical network flows, normal traffic usually dominates intrusion traffic in terms of volume [2]. Nevertheless, certain types of intrusion attacks are more vulnerable and

impactful, leading to increased logging activity and network events [3]. An intrusion detection system (IDS) is considered a potent defense mechanism, capable of mitigating sophisticated threats and attacks [4]. Traditional systems often struggle to keep pace with emerging threats and the continuous growth of cyber attacks [5], and they may face difficulties due to uneven traffic distribution and high-speed data movement [6]. Another significant challenge is the difficulty in accurately labeling traffic datasets, which is essential for the development and evaluation of effective intrusion detection models [7]. In this case, some of the methods have been developed to solve the existing issues by deploying characterization concept [8] and the kernel algorithms [9],

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and some are designed for tackling the complicated uncertainties [10] and fractional differentiating challenges [11]. But, these models have the complexity in solving the issue by taking lesser periods. Thus, deep learning models with optimization algorithms act as an intelligent model for offering a unified system to solve these uncertainties and complicated issues. The deep learning models help to enhance smart grid reliability and efficiency [12], while the optimization algorithms, such as the differential evolution algorithm [13], offer adaptability and improved global and local search capabilities. Together, these models prove their unique features in offering comparable effectiveness to other models.

Advantages and Disadvantages of the Existing Network ID Approaches: Traditional ID systems are classified into three major groups, and they are signature-aided, anomaly-aided, and hybridized approaches. Here, the models based on signatures include Suricata [14] and Snort [15], which are effective in identifying assaults by providing high accuracy and reduced false positives. In recent days, they have been successfully employed in ID, as they can handle large datasets more effectively than various machine learning techniques [16]. They are potential choices in the recent cyber environment, as the training allows scalability and adaptability to evolving network threats. However, in IDS datasets, the complexity of analyzing the data patterns and anomalies in the network traffic adds challenges to the feature engineering [17]. Moreover, they cannot detect novel or zero-day attacks and require constant updates of signatures and often fail to detect false negatives and have bottleneck issues while training large data [18]. Overfitting is a major concern in NID, and it is caused by ineffective training processes. Therefore, deep learning models, including Convolution Neural Networks (CNN) [19], Recurrent Neural Networks (RNN) [20], and Graph-based techniques (GNN) [21], have been developed as these perform well in achieving generalization performance; however, these models face the gradient explosion problem. Moreover, the methods centered on the anomaly involve Autoencoder [22], One-Class Support Vector Machine (SVM) [23], and Practical Pretrained Spatio-Temporal Graph Neural Network for Network (PPT-GNN) [24]. These models have the ability to identify the unseen threats and assaults by leveraging the modeling of normal system actions. Although these are adaptable, they have issues in terms of noise sensitivity and false positives. Finally, the hybrid approaches, such as Convolutional Neural Network with Long-Short Term Memory (CNN + LSTM) [25] and Recurrent Neural Network and Gated Recurrent Units (RNN-GRU) [26], performed by combining the power of each individual model, thus enhancing the coverage, along with the accuracy of detection. However, these approaches suffer from computational overhead and require more training data, which limits their real-world applicability.

Moreover, some of a recent deep learning methodologies, like GNN, has increased their diagnosis accurateness by representing the data in a graph-based format, and thus have better representation of the complex features. Still, the graph-based techniques are not able to understand the interdependencies between the data and intrusions, leading to false positives [27]. Attention mechanisms [28] have been added to the deep learning architecture to create a better focus on the significant features. It also helps in capturing the edge features and leads to better ID accuracy [29]. Some works have focused on creating adversarial samples for the ID, but this mostly leads to the requirement of high memory. In [30], a fuzzy-based approach was used with the deep learning network, and an optimizer was involved in feature engineering, but defining different fuzzy rules leads to complexity in training. Even though a lot of deep learning focuses on improving detection accuracy, the necessity of proper feature engineering is necessary. So, many appropriate feature fusion methods have been developed to overcome the class imbalance issue, and better learning of significant features representing the normal and intrusion traffic [31]. Yet, they have some research gaps, even with enhanced systems effective in managing the class imbalance issues, along with the feature representation.

Research Gaps and Need for New Model: Although the existing

research presents numerous models for ID, there exist some research gaps that need to be addressed. Initially, the traditional models are not effective in combining various feature types, such as deep, statistical, and embedding data into a single and accurate set of feature representation. Then, the traditional systems only rarely optimized the related contribution of extracted features, resulting in suboptimal performance when detecting the network intrusion, which also increases the cost. Finally, the adaptability of the traditional systems is poor when considering the more complicated assault patterns and evolving threats, which limits the model's robustness in varying and dynamic environments. Thus, it is crucial to propose an innovative method to address all these gaps.

Research Contributions: To address these gaps, this research concentrates on creating a balanced training approach by defining an appropriate feature fusion process for deep learning. Deep learning is designed to achieve high detection accuracy through a compact and intelligent design. The contributions of the proposed ID model employing deep learning are given below.

- To implement an ID model with a feature fusion strategy that can identify and distinguish abnormal behaviors in network traffic. By leveraging a hybrid deep learning network system, the developed system aims to accurately detect malicious activities deviating from typical network patterns, ensuring enhanced security. The system ensures accurate detection, enabling rapid responses to safeguard the network from potential cyber threats.
- To extinguish a weighted feature fusion procedure by combining different characteristics obtained from different sources, including statistical features, deep learning, and t-SNE-related features. The fusion process is optimized using the developed RCMFA, which fine-tunes a weights assigned to each characteristic, ensuring most relevant information is emphasized. This ensures the model achieves generalizability under unseen datasets, thus making it suitable for real-time performance.
- To develop an RCMFA optimizer to increase the correlation with the target intrusion detection task while minimizing inappropriate or superfluous characteristics. This optimization significantly diminishes a dimensionality of an input data, which helps to alleviate the curse of dimensionality and reduces computational complexity and cost. By using RCMFA, the system can better focus on the most informative features, enhancing an overall efficacy and accurateness of the diagnosis process.
- To propose an MDDHN-AM deep learning model as a hybrid structure to engage in intrusion detection. The model can handle sequences of different lengths and is also applicable for time series information like the ID dataset, thus enhancing the robustness even with complicated traffic patterns. Hybridization of TCNN and RNN through activation enables the system to handle large dataset sequences, and also achieve high speed and precision in the diagnosis of intrusions in the system.

Benefits and Advancement of Knowledge: A developed research work is effective in IDS sectors as it has the ability to offer adaptive and scalable results when managing the imbalanced, along with the high-dimensional network. It also enhances its effectiveness in detecting both the unknown and known network threats by using the weighted feature fusion with the hybrid deep learning framework, thus minimizes the computational overhead and complexity. By providing enhanced security and robust processes, the developed system helps the researchers, which balances the adaptability, accuracy, and efficiency, thus enhancing the development of next-generation IDS.

"The organization of the research is done as follows: The traditional ID model's challenges and advantages are described in Section II. Details related to the dataset, developed system, and implemented methodology is specified in Section III. The feature extraction details and optimally tuned weighted feature fusion explanations are given in Section IV. The

developed ID technique is briefed in Section V. The experimental setup and outcomes of the developed ID system are specified in Section VI. The conclusion and research limitations are provided in Section VII."

2. Literature survey

2.1. Related works

In 2019, Otoum *et al.* [18] investigated a cluster-based intrusion prediction model employing deep learning techniques to detect network intrusions. According to the numerical outcomes, the implemented ID model achieved high accuracy and could identify different malicious activities within the network. In 2020, Zhong *et al.* [19] proposed an ID model based on the big data concept, where network traffic information was presented at the payload level. Both content and behavioral features were extracted from the traffic data. Compared to earlier systems, this strategy improved intrusion prediction rates and handled large volumes of traffic.

In 2021, Wang *et al.* [20] utilized a Deep Belief Network (DBN), which could adapt its depth according to incoming data. To intercept a network from dropping into local optima due to high complexity, the training process considered fixed thresholds and weights. This approach enabled the model to operate in high-traffic and complex network scenarios. In 2019, Khan *et al.* [34] designed a deep network combining a stacked autoencoder and softmax technique for effective intrusion detection. The framework consisted of two phases: classification and detection. In the classification phase, the prediction probability scores determined network traffic, which was then used in the detection phase to identify various attacks.

In 2021, Ashiku *et al.* [35] developed a novel framework to implement a reliable and adaptable ID system capable of detecting system threats. The model focused on enabling flexible predictions with learning capabilities to identify recognized behavioral patterns. A real-time dataset was constructed to assess system performance. In 2022, Ahmed *et al.* [36] introduced an entropy-based, packet-level ID system. A load balancing mechanism monitored incoming traffic, and the network enhanced presentation regarding entropy metrics and pooling efficiency.

In 2021, Wang *et al.* [37] proposed a hybrid deep learning framework for ID, combining a random forest for detection and classification with feature extraction to enhance both accuracy and efficiency. An evaluation demonstrated best presentation than traditional models; however, it struggled with large traffic volumes. In 2020, Wang *et al.* [38] developed a multi-scale convolutional model for detecting malicious activity, using various convolution kernel sizes to extract features at multiple levels. The results demonstrated high accuracy and True Positive Rate (TPR).

In 2024, Maddu *et al.* [39] presented a deep learning-based intrusion detection approach addressing data imbalance using Deep Convolutional Generative Adversarial Networks (DCGAN). Adversarial training improved generalization, which makes it applicable for high-speed environments. In 2024, Aljehane *et al.* [40] introduced a Gravitational Search Algorithm-based feature selection combined with an Optimal Quantum Neural Network (GSAFS-OQNN) for intrusion detection and classification. The preprocessing step ensured clean data, and the optimizer effectively selected relevant features without increasing network complexity.

In 2024, Aljehane *et al.* [41] used the Golden Jackal Optimization Algorithm-assisted deep learning model for network intrusion detection. The model effectively identified intrusion types, ensuring network security. In 2024, Saleh *et al.* [42] employed a combination of Gaussian Naive Bayes (GNB) and Stochastic Gradient Descent (SGD) techniques to detect network intrusions in WSNs, addressing specific detection challenges. In 2024, Karthikeyan *et al.* [43] applied a machine learning approach with the Firefly Algorithm to achieve high-accuracy intrusion detection, particularly suitable for critical infrastructure and industrial automation.

In 2023, Ullah *et al.* [44] introduced a Multi-head Attention-based Gated Recurrent Unit (MAGRU) model for detecting malicious activities. Multi-head attention improved learning capability, while GRU captured network behavior patterns. The model was estimated on "Edge-IIoTset and MQTTset datasets". Finally, in 2024, Mirlashari *et al.* [45] developed an E-GraphSAGE network using Graph Neural Networks (GNNs) to address emerging cybersecurity challenges. Unlike earlier models, this system employed an updated message-passing operation that merged node and edge features through transformations, enabling effective network-wide detection.

In 2025, Khatami *et al.* [46] have developed a Five-Dimensional Gray wolf optimizer (5DGWO), combined with an autoencoder for detecting the intrusion. The developed model attained above 94 % of accuracy when compared with others, thus proving a promising, scalable, and efficient solution for IoT security. In 2025, Abualhaj *et al.* [47] have proposed an ID model using the power of the Whale Optimization Algorithm (WOA). It achieved 98.76 % of recall, 99 % of accuracy, and 98 % of precision. Thus, the outcomes prove that a proposed method was more intelligent and suitable for reducing the class imbalance issue. Finally, Ogunseyi *et al.* [48] have proposed an LSTM model, learning the explainable Artificial Intelligence (XAI), and the performance was enhanced by the incorporation of the Firefly Algorithm (FA). This model provided valuable insights into the future network.

Recent Works on Class Imbalance Issues: Class imbalance remains one of the greatest challenges in NID research. Recent works addressing class imbalance have focused on improving network efficiency in handling large datasets and complex attack scenarios. In [49], a framework based on embedding logic was proposed to handle hierarchical dependency and class imbalance. It addressed the issue by accessing high-level synthesized features and mutating important features, but suffered from high training complexity. In [50], class imbalance and feature redundancy were tackled using a self-supervised architecture. In [51], a dual attention-based structure was designed, which addressed class imbalance by increasing attack traffic via traditional data augmentation techniques; however, this approach risked losing higher-level features. In [52], an XGBoost-based network was introduced to identify minority attack classes, achieving high precision and recall. Moreover, in [53], XAI with XGBoost was developed, which was effective in addressing the class imbalance issue, and finally, to prove the generalization ability, DenseNet [54] structure was designed by Mahmud *et al.* in 2024.

Deep Learning-Based RL Techniques: Reinforcement learning (RL) techniques are particularly well-suited for large datasets, especially sequential data. Most RL techniques applied to NID have demonstrated improved performance because of their capability to manage noisy information. In [55], a deep RL model was developed to detect various intrusion attacks, although the architecture faced complexity issues because of the adaptive nature of attacks. In [56], a Transferable and Adaptable NID System (TA-NIDS) was proposed, overcoming challenges posed by adaptive attack types. In [57], a Q-learning-based RL model (Q-DRL) was introduced, which effectively handled sequential information and enabled faster response times. In [58], a Micro Reinforcement Learning Architecture (MRLC) was designed for small-scale NID systems, addressing computational cost concerns.

2.2. Problem statement

Due to digitalization, many applications related to communication services are in high demand. Most private services offer their services on demand, such as cloud computing, allowing authorized users to access them. However, malicious activities often target these networks, aiming to disrupt normal traffic and engage in financial crimes. Therefore, NID systems are increasingly critical, as they alert the system to incoming malicious activity before it causes significant damage. Since networks typically handle large volumes of data, the proportion of malicious activity is comparatively small, leading to substantial data imbalance,

Table 1
Advantages and disadvantages of traditional network ID approaches.

Authors' [citation]	Methods	Advantages	Disadvantages
Otoum et al. [18]	RBC-IDS	- It can learn the probability distributions over an input set. - It automatically detects attacks from data patterns using static methods.	- It can only be used in smaller networks. - It is difficult to determine the energy gradient functions.
Zhong et al. [32]	BDHDLs	- It enhances the intrusive attack rates. - Low design cost. - High-dimensional intrusion features are obtained for training. - Find a stable and optimal solution.	It does not provide an optimal solution.
Wang et al. [33]	DBN	- High robustness. - Less consumption time to execute the large data.	Leads to large training time under complex conditions.
Khan et al. [34]	TSDL	Exposures and cybersecurity vulnerabilities can be effectively detected.	Does not undertake multi-tasking.
Ashiku and Dagli [35]	DNN	Better handles the incoming data without a bottleneck.	It has the minimum capacity to manage zero-day attacks.
Ahmed et al. [36]	SDN load-balancing model	Noisy data are better handled.	Active learning mechanisms are not used.
Wang et al. [37]	SAE-RF		- Real-time traffic detection is not achieved. - Minority class samples are rejected during training, leading to class imbalance.
Wang et al. [38]	DMCNN	- Generalizes to new data. - Reduces false alarms.	It cannot detect data misuse on real-time streaming data.
Maddu et al. [39]	DCGAN, ResNet152V2	Handles large data with less training time.	The robustness of this approach is low.
Aljehane et al. [40]	GSAFS-OQNN	It detects various types of cyber threats in the system.	- It is not suited for huge-scale networks. - A responsiveness of this approach is minimum.
Aljehane et al. [41]	GJOADL-IDSNS	High accurateness in diagnosing different attacks.	A convergence of a system is low.
Saleh et al. [42]	GNB and SGD	Ensures better data flow in the network.	Creates unnecessary alarm.
Karthikeyan et al. [43]	FA-ML	Optimizer handled better features for training.	Overfitting issues may occur.

especially when using machine learning or deep learning strategies during NID design. Although much research has addressed NID challenges such as data imbalance, precision, and false positives (as listed in Table 1), issues like overfitting and generalization still require further improvement.

- Network traffic data possesses a multitude of characteristics, making existing models struggle to extract meaningful patterns, resulting in higher computational complexity and longer processing times. Deep

learning models address this challenge by automatically reducing dimensionality, learning hierarchical representations, and filtering out irrelevant features while preserving critical ones. By extracting abstract features at each layer, deep learning models help mitigate the curse of dimensionality.

- Raw network data often contains irrelevant or redundant features. Selecting the most relevant features is essential to improve diagnosis accurateness and decrease computational load. Existing models often fail to perform proper feature engineering, which is key to resolving overfitting issues. This challenge can be addressed through feature fusion, which identifies the most significant attributes in the dataset. Feature fusion retrieves meaningful features that capture the most discriminative patterns, reducing dimensionality while preserving essential information for effective intrusion detection. This leads to improved system performance and reduced processing time.
- Timely detection is critical for mitigating cyber threats. Traditional IDS often struggle to process a large volume of information in real-time, causing delays in detecting and responding to intrusions. This issue can be resolved using a hybrid detection model that combines two different networks to detain temporal dependencies and spatial features from system traffic. Such a combination ensures the system can manage large data in real-time while handling large diagnosis accurateness, providing fast alerts to identify and mitigate intrusions.
- Existing ID systems often experience high false positive and false negative rates, which reduce their effectiveness. False positives trigger unnecessary alarms, while false negatives may miss intrusions. Weighted feature fusion can mitigate this issue by extracting multiple features using different techniques and assigning them weights based on relevance. Optimizing these feature weights permits the network to prioritize a more discriminative characteristics, reducing false negatives and positives and improving the method's accurateness in identifying legitimate attacks.
- As cyberattacks continuously evolve, IDS must adapt to detect new and previously unknown attack patterns. Traditional signature-based systems often fail to recognize novel threats. Optimization algorithms can address this challenge by dynamically adjusting system parameters, such as feature weights and model thresholds. Through continuous learning and optimization, the IDS becomes more adaptable, capable of detecting emerging attack strategies, ensuring continued effectiveness against new threats.

3. Systematic illustration of proposed ID scheme with ID dataset acquisition

3.1. Description of proposed scheme

Recent works in deep learning-related NID focus on mitigating class imbalance issues and enhancing the method's generalization to untapped information. However, most existing works emphasize building high-speed networks or increasing network depth to handle large incoming data streams. In most cases, the resulting ID dataset suffers from class imbalance, as the malicious class is comparatively smaller than normal traffic. Most existing networks face challenges from class imbalance during training, which often results in high false positives and false alarms. Recently, deep learning models have employed various intelligent strategies to handle large network traffic and improve detection accuracy. Feature interdependencies play a key role in enhancing detection, allowing better identification of malicious and normal traffic. This research develops a hybrid deep learning model with a novel feature fusion technique to achieve high robustness on large datasets and improved detection of different malicious attacks in the network. Fig. 1 provides the diagrammatic design of the offered ID framework using deep learning.

"The newly designed ID system is used to predict anomalous activities in the networks and thereby enhances security. The network data

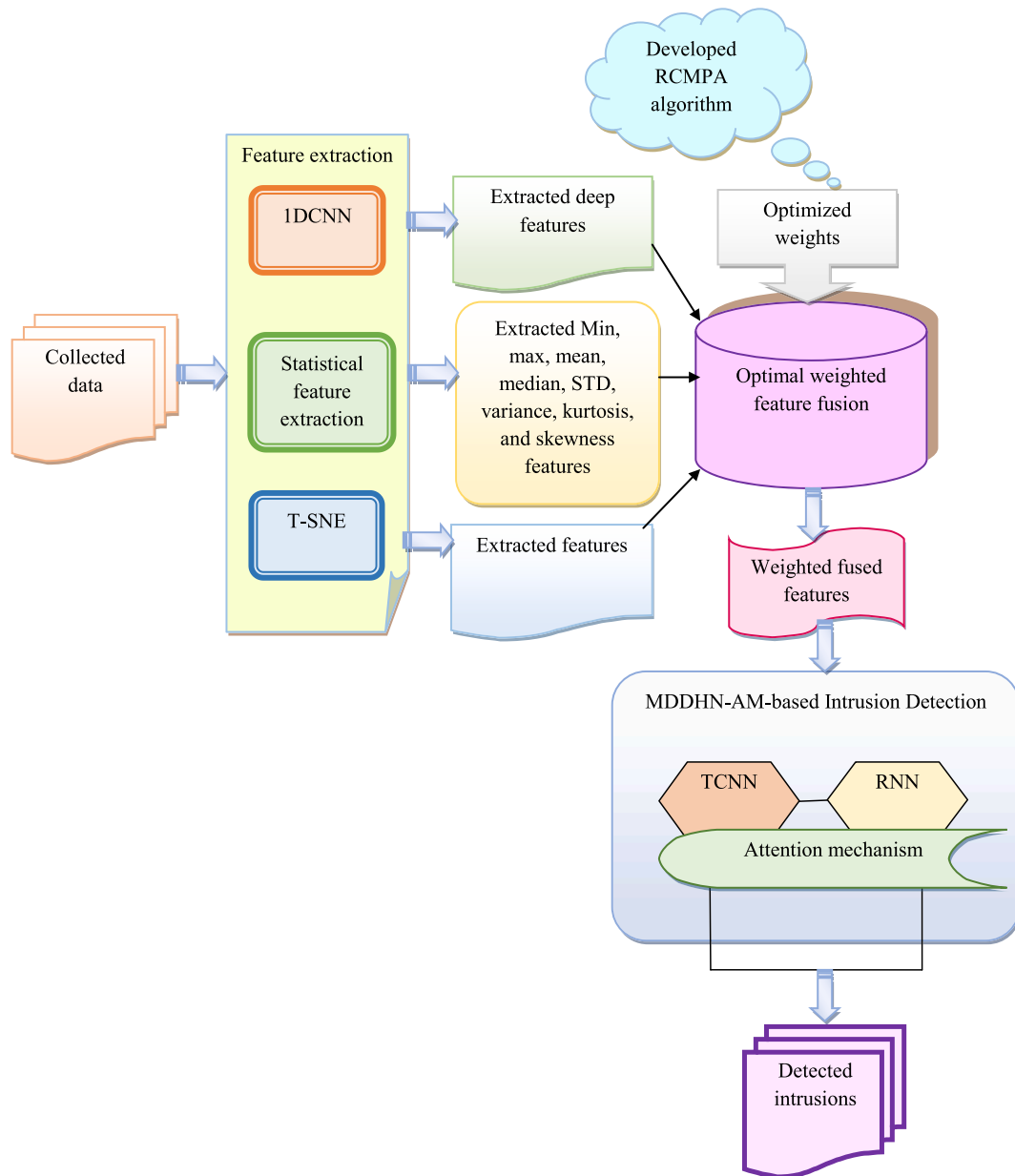


Fig. 1. Diagrammatic design of an offered ID framework using deep learning.

related to different activities is gathered from standard datasets. The data is then processed by the developed weighted feature fusion model, where important statistical, deep, and temporal features are identified. A 1DCNN network is employed to capture the deep features, and it effectively captures time-series features arriving at different intervals. Statistical parameters such as min, max, median, standard deviation, variance, kurtosis, and skewness are extracted, as they improve the network's ability to detect zero-day attacks and help the network adapt to various threats. Next, the T-SNE-based features are obtained, which assist in managing high-dimensional data. Each extracted feature is then set for fusion based on optimal weights selected by the developed RCMPA algorithm. The RCMPA tunes the weights to maximize correlation among features. This weighted feature fusion ensures that the network captures interdependencies among features and allows it to generalize to new data. Additionally, by identifying the most significant features, the training time is significantly reduced. Then, the weight-optimized fused features are given to the developed MDDHN-AM to perform the final intrusion detection process, resulting in enhanced

performance when detecting the malicious activity in the network. The proposed MDDHN-AM is the combination of TCNN and RNN, and an attention mechanism is also used with it to focus only on the essential features, enabling high detection accuracy and faster computation. Finally, the developed model provided an accurate intrusion detection outcome, leading to high robustness and precise malicious activity detection results."

3.2. ID dataset collection

"Dataset-1 (APA-DDoS Dataset):"

"<https://www.kaggle.com/datasets/yashwanthkumbam/apaddo-s-dataset>. It contains three files in a CSV and PCAP format, and the data contains twenty-three columns. Attack patterns like PUSH-ACK and ACK attacks are briefed under this dataset with a total size of 151200×19 . The dataset includes key attributes for network traffic analysis and intrusion detection. The features cover both source as well as target IP addresses (ip. src, ip. dst) and TCP ports (tcp.srcport, tcp.dstport) help in

identifying communication endpoints. The IP protocol version (ip.proto) indicates the protocol type, while frame length (frame.len) measures packet size. TCP flags (tcp.flags.syn, tcp.flags.reset, tcp.flags.push, tcp.flags.ack) show connection control and data transfer states, while IP flags (ip.flags.mf, ip.flags.df, ip.flags.rb) reveal information about packet fragmentation. Additionally, TCP sequence and acknowledgement numbers (tcp.seq, tcp.ack) track packet order and receipt, and frame time (frame.time) timestamps of each packet. The dataset also includes packet and byte counts, such as total packets (Packets) and bytes (Bytes), as well as transmitted (Tx Packets, Tx Bytes) and received packets (Rx Packets, Rx Bytes). These attributes are important for monitoring network behavior and identifying anomalies or potential intrusions.”.

“Dataset-2 (WUSTL-IIOT): <https://www.cse.wustl.edu/~jain/iio/index.html>. The information on file is presented in a CSV. The “0” value is considered for the normal traffic, and the “1” value is considered for the attack traffic in the target column. This dataset contains 7 columns representing total packets, source port, total bytes, destination packets, source bytes, and source packets. The size of the dataset is 7037983×6 . The dataset includes key attributes for analyzing network transactions. Source Port (Sport) identifies the source application’s port, while Total Packets (TotPkts) and Total Bytes (TotBytes) measure the volume of data exchanged. Source Packets (SrcPkts) and Destination Packets (DstPkts) track packet flow between source and destination, and Source Bytes (SrcBytes) and Destination Bytes (DstBytes) capture the data sent and received.”.

“Dataset-3 NSL-KDD: <https://www.kaggle.com/datasets/hassan06/nsllkdd>”: This dataset contains different intrusion information collected under 21 different classes, and it does not have redundant data records. This dataset can be trained and tested according to the severity levels of attacks, and thus can be used in cybersecurity-based research.

“Dataset-4 UNSW-NB15: <https://www.kaggle.com/datasets/dhoogla/unswnb15>”. This has related information for the intrusion detection, and offers clean data, as the pre-processed data are free from noisy features. It has 4 different files representing the original, clean, reorganised, and updated file sequences.

Dataset-5 AWID-CLS-R: <https://www.kaggle.com/datasets/zhigongcui/awidclsr>. This dataset collects the intrusion data from the network and contains more than 150 related attributes. It represents the flooding, impersonation, and injection type attacks.

The gathered ID information is specified by P_i^K , where, $k = 1, 2, \dots, K$, and the total data is denoted by K . Finally, from the total data, 70 % of the data is utilized for training, 15 % is for testing, and the remaining 15 % is for validation. These splits were performed randomly, instead of using any techniques, like cross-validation and time-based split.

3.3. Implemented RCMPA

Optimization algorithms are largely beneficial in obtaining the optimal solutions without increasing a computational complexity of the network. In high-end applications like NID, the data dimension can lead to large resource requirements, and thus, the research needs to find ways to detect intrusions with lower complexity. Also, the features in the ID dataset are redundant and contain a lot of irrelevant data. Using optimization algorithms in feature engineering provides better insight into finding the significant features, and it also improves an efficiency of intrusion diagnosis. In this work, the RCMPA optimization algorithm is newly developed from CMPA [59], and it is utilized to perform weight optimization and to create a weighted fused feature set.

Reason for CMPA: CMPA gives high flexibility for coding, and it is cost-effective. Unlike traditional optimization algorithms that might get stuck in specific regions of the search space or fail to adapt to changing patterns, CMPA provides a more robust and efficient solution, enhancing the overall functionality of ID models. However, the convergence rate of CMPA requires further improvement, as the chances for a solution are

still vulnerable to local traps. In certain cases, the existing CMPA algorithm faces instability in its exploration and exploitation capabilities. Hence, the newly developed RCMPA optimization is implemented to resolve the above issues.

Justification for the Terminology: The developed optimization algorithm, named RCMPA, is inspired by the mask prevention mechanism applied during the COVID-19 pandemic. This motivates to development of an enhanced and explorative population protection optimization algorithm. Here, the concept of mask protection is changed into an optimization mechanism, which aids in protecting the optimal solution by using the exploration ability to new regions, thus avoiding the local optimum, while enhancing the speed of convergence. Thus, this terminology, RCMPA, highlights the innovative algorithm underlying the optimization process.

Novelty: The static approach in existing CMPA leads to premature convergence or an inability to further explore the solution space effectively, especially in complex and high-dimensional problems. The developed RCMPA addresses these limitations by dynamically calculating the random parameter rnd , depends on a relative fitness score of a current solution $CurFit$, a best solution $BesFit$ found so far, and the worst solution $WorFit$ encountered during a search. To calculate the formula for random parameter is adaptive, taking into account the fitness progress made at each step, which helps the algorithm adjust its behavior depending on the present state of the optimization process. The random variable in the existing CMPA is denoted by rnd and it is chosen in the interval of $[0, 2]$. In the developed RCMPA optimization, the random parameter rnd is calculated utilizing the adaptive formula as in Eq. (1).

$$rnd = - \frac{(BesFit * CurFit)}{(WorFit * CurFit)} \quad (1)$$

The developed RCMPA optimization technique introduces a key innovation by adapting the random parameter used during the optimization process, providing a more efficient search strategy compared to conventional CMPA. The primary advantage of this adaptive random parameter is that it permits the RCMPA algorithm to more efficiently inspect the search space. By adjusting the random variable based on the fitness values, the RCMPA is better equipped to avoid premature convergence, a common issue in many optimization algorithms, and can better navigate the complex feature space to determine the optimal weights for fusing the feature set. This adaptability makes the RCMPA particularly effective in scenarios where the optimization problem is highly dynamic, such as intrusion diagnosis, where the attack patterns and nature can convert over time.

The adaptive nature of RCMPA means that the random variable is not fixed within a predefined range but evolves as the algorithm progresses, allowing for more flexibility in the search process. In the early stages of the search, when solutions are still far from optimal, the algorithm uses a broader search strategy, encouraging exploration of different regions of the solution space. As the algorithm identifies better solutions, the random parameter becomes more focused, promoting exploitation of promising areas while still maintaining some exploration to ignore getting adhere in local minima. This balance between exploitation and exploration is key for efficiently navigating complex solution spaces, as seen in intrusion detection systems, where the data patterns and attack strategies can dynamically change.

Performance of the Developed RCMPA Compared to Standard Optimizers: Compared to existing optimization techniques like GA [59] and PSO [60], the developed RCMPA offers several advantages. Traditional algorithms use static random variables or velocity adjustments, which can make the search process rigid and less responsive to the needs of the optimization problem. In contrast, the adaptive random parameter in RCMPA allows an algorithm to adjust its search behavior continuously, enhancing its ability to balance a trade-off between (refining existing solutions) and exploration (searching new regions) exploitation. This capability makes RCMPA more effective in handling complex, evolving

datasets like those encountered in cybersecurity tasks. The ability to dynamically adjust the search process confirms that an algorithm does not waste resources exploring unpromising regions, while still being able to escape from local optima and discover better solutions. In this sense, RCMPA is better suited for intrusion detection, where it is critical to continuously adapt to new attack patterns and the changing nature of network traffic. This makes RCMPA a more robust, flexible, and effective optimization technique for improving feature fusion and, ultimately, the ID system's performance. The primary advantage of this dynamic adjustment is that RCMPA avoids the common problem of premature convergence, where the algorithm might settle on suboptimal solutions too early in the search process. By continually adapting the random variable based on the fitness landscape, RCMPA can more effectively explore the search space and refine the solutions as it progresses, which significantly enhances its convergence toward the global optimum or a near-optimal solution. This adaptability is especially valuable in intrusion diagnosis, where the information is noisy, high-dimensional, and subject to change as new attack strategies emerge. The developed RCMPA's flowchart for finding optimally tuned weights for the feature fusion is displayed in Fig. 2, and the pseudocode is explained in Algorithm 1.

Algorithm 1: Pseudocode for finding optimally tuned weights for the feature fusion

```

Begin
Initialize the population of  $N$  candidate weight vectors  $KL_w^{Op}$  randomly within
 $[W_{min}, W_{max}]$ 
For each weight vector  $KL_w^{Op}$  in the population:
Compute the fused feature vector.
Evaluate fitness
Identify the best feature with the highest fitness.
Iteration = 1
While Iteration ≤ MaxIteration
For each candidate weight vector:
Update using RCMPA operators (Mask Protection inspired)
Protect promising weights (high fitness)
Explore new candidate weights for low-fitness vectors.
Enforce bounds
Compute the weighted fused feature.
Evaluate fitness function
Update the best weight if a new weight vector achieves a higher fitness.
Iteration = Iteration + 1
Construct the final tuned weighted feature vector.
Return Optimal weight
End

```

4. Extraction of features and optimal tuned weighted feature fusion for detecting intrusions

4.1. 1DCNN-based deep feature extraction

The input data P_i^K are fed into the 1DCNN [61] network and which helps extract the deep features from the ID dataset.

“Motivation for 1DCNN: The conventional models used for feature extraction are not effective in capturing the subtle changes and temporal correlations, particularly when concerning high-volume contexts. However, the 1DCNN is more preferred as it is intelligent in learning the hierarchical, as well as the local patterns, automatically without the need for manual feature engineering. It enables the proposed model to extract the high and low-level features, which help encode both the short and long-term dependencies present in the system traffic data. This aspect is essential to provide accurate intrusion detection and handle the issue of class imbalance.

Process: In the context of ID, 1DCNN-based deep feature extraction is a powerful approach that learns hierarchical representations by applying convolutional layers to capture spatial and temporal patterns. These layers help the network focus on discriminative features that are relevant for identifying intrusions, making the model more efficient. Under each layer of processing, the dimension of features is reduced, leading to a better feature extraction strategy. It ensures the loss of

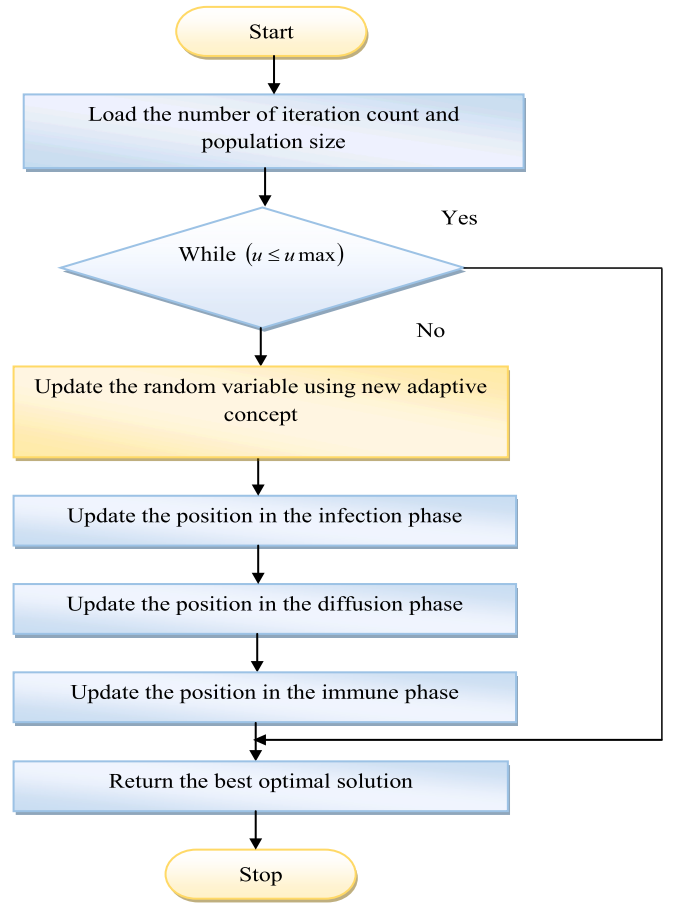


Fig. 2. Flowchart of a designed RCMPA optimization.

significant information by mapping the features from one layer to another. The layers with pooling and convolution functions are controlled through a 1-D convolution unit. It extracts the selected information from a one-dimensional layer by utilizing the shared weights and locally accessible area from the deep network.”.

Convolution layer: The convolution layer in the 1DCNN is used to extract the initial set of deep features from raw data, and its layers are represented in Eq. (2).

$$b_l^n = h \left(\sum_{k \in O_l} b_k^{n-1} * M_{kl}^n + d_l^n \right) \quad (2)$$

Here, the term b_l^n is the l^{th} neuron outcome in layer n , and a filter or weight matrix that convolves with the input from the previous layer is M_{kl}^n . The terms O_l , n , d_l^n indicate the input value, total layer count, and bias. The activation function used for training is represented by h , and it may be a sigmoid or hyperbolic tangent function.

Fully Convolution: The convolution layers are followed by a full convolution unit, and it is attached to a pooling unit to reduce the over-sampling, and its final output is calculated by Eq. (3).

$$U_l^n = \text{Max}(c_k^n), \quad k \in T_k^n \quad (3)$$

Here, the 1DCNN's pooling layer is noted by T_l^n . The term $u \times 1$ is the size of the kernel. The pooling layer's output feature is noted by c_k^n . The term h_x is the feature map with a cascaded function. The value Q is calculated by Eq. (4).

$$Q = h(d_q + v_q h_x) \quad (4)$$

Here, the values d_q and v_q are denoted as weight and bias, respectively. The output of extracted parameters is indicated by E_f^d .

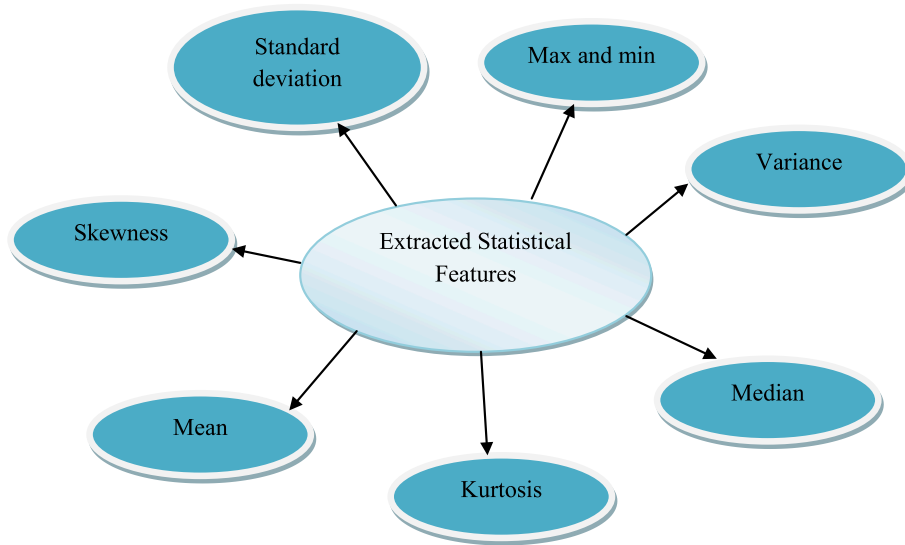


Fig. 3. Diagrammatic View of the Extracted Statistical Features.

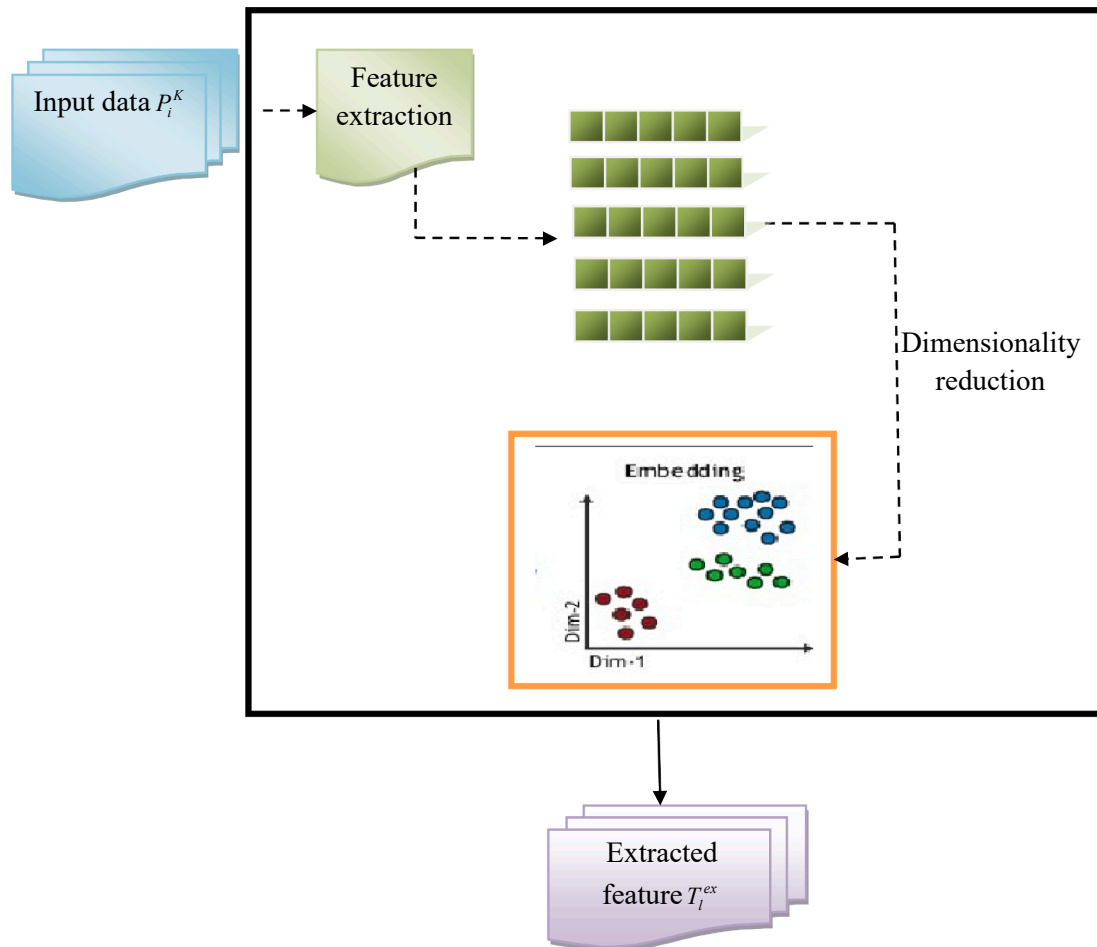


Fig. 4. Pictorial View of the TSNE-based Feature Extraction Model".

4.2. Statistical feature extraction

The input data P_i^K is first provided for statistical feature extraction. Statistical features are important because they help to identify new attack types and make the network more adaptable through their sta-

tistical properties. For example, an attack such as a DDoS attack may lead to sudden spikes in the number of packets or traffic volume over a short period, which can be detected using statistical features such as mean and variance. On the other hand, network scans or probing attacks might exhibit patterns in the frequency and distribution of packets, which are captured using higher-order statistical moments. In the sta-

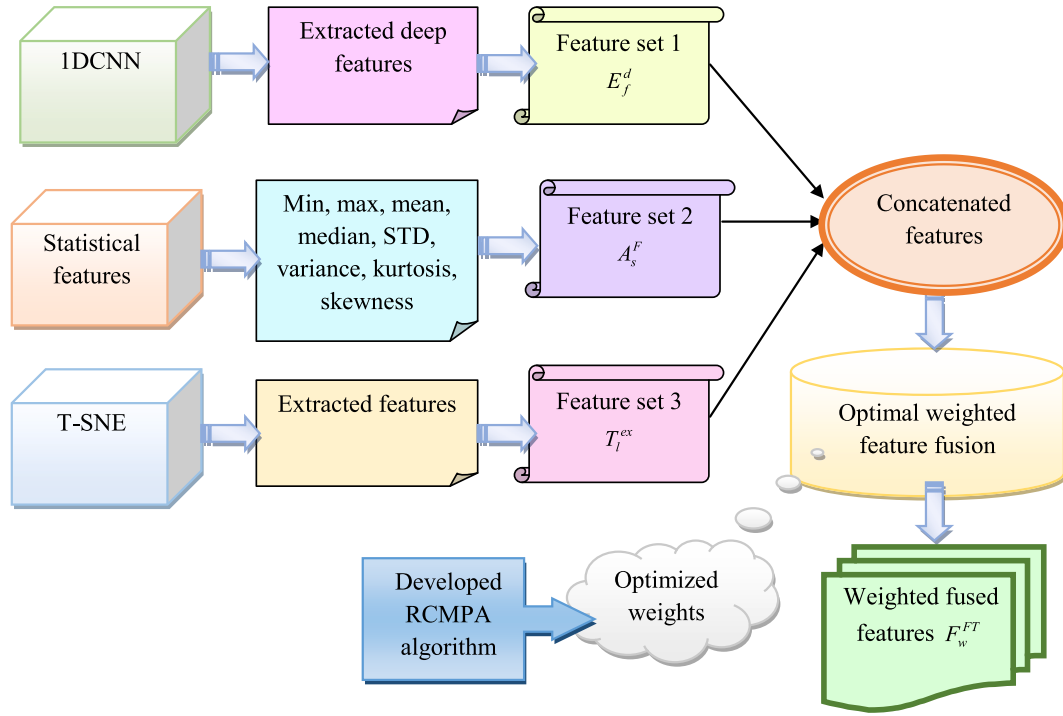


Fig. 5. Structural design of the tuned weighted feature fusion using RCMPA.

tistical feature extraction phase, standard deviation, max, variance, min, median, kurtosis, mean, and skewness are extracted from the raw data. Fig. 3 offers the diagrammatic view of the extracted statistical features.

4.3. TSNE-based feature extraction

The input data P_i^K is next provided to the T-SNE [62] model. T-SNE-based feature extraction is particularly useful for detecting intricate relationships between features that are not easily captured by traditional methods. For example, network traffic can exhibit non-linear relationships, where certain patterns or anomalies are more complex and difficult to interpret using basic statistical methods. By decreasing the dimensionality of an information, T-SNE makes it easier to visualize complex patterns, making it a powerful tool for recognizing hidden intrusions or abnormal behaviors in network traffic. An T-SNE algorithm involves two main steps. In a high-dimensional space, probability diffusion is carried out as the initial stage. In contrast, there is a lower chance of choosing two different objects in the lower-dimensional space. To calculate similarity distances, different methods employ various criteria. The similarity distances between two items are computed using conditional probability. A extracted T-SNE characteristics are noted by T_l^{ex} . Pictorial view of a TSNE-based characteristic removal model is given in Fig. 4.

4.4. Tuned weighted optimal features

The extracted deep features from 1DCNN E_f^d , statistical features A_s^F and T-SNE features T_l^{ex} are fused in the tuned weighted feature fusion. In IDS, feature selection and fusion are critical steps, and thus, this work uses RCMPA to achieve optimized feature selection and fusion, which determines the optimal weights for various characteristics depends on their pertinence and contribution to a detection process. Dataset 1 contains a total of 16 features, and Dataset 2 has 6 features representing different network and attack conditions. During 1DCNN processing, a total of 679 characteristics are collected from each dataset, and likely, T-SNE results in 679 features. The statistical features account for 12 feature sets. Processing this large feature set leads to high training time.

Therefore, the RCMPA algorithm selects optimal weights to choose 10 appropriate features from each Dataset 1 and Dataset 2. The final 10 features from each dataset are used for training the ID detection model.

Role of Novel RCMPA in Feature Fusion: The proposed RCMPA works by iteratively selecting and tuning feature weights, based on the correlations between features and their effectiveness in detecting intrusions. It analyzes the features and assigns weights to them, favoring those with higher relevance to the detection process while diminishing the impact of less relevant features. This approach helps in addressing the dimensionality problem while training the large network traffic data, where a large number of characteristics may leads to overfitting. By assigning optimal weights to the most relevant features, RCMPA enhances the IDS's capability to faithfully classify normal and malign traffic with high accuracy. An implemented RCMPA optimization is employed to optimize the weights accordingly to maximization of the correlation coefficient. The extracted features, such as deep features from 1DCNN E_f^d , statistical features A_s^F and T-SNE features T_l^{ex} are combined and multiplied by optimized weights to get the final tuned weighted fused feature set. The features are added and multiplied by optimized weights to get the tuned weighted feature. The final output representing the tuned weighted fused feature is given $F_w^{FT} = (E_f^d + A_s^F + T_l^{ex}) * KL_w^{Op}$, where the term KL_w^{Op} is the optimized weight from the novel RCMPA algorithm. The extracted deep feature in the 1DCNN is indicated by E_f^d . The extracted statistical feature is noted by A_s^F . The extracted T-SNE feature is noted by T_l^{ex} . The final tuned weighted feature is noted by F_w^{FT} .

The correlation coefficient is a statistical calculates for finding the relation between any two parameters. The objective function representing the correlation coefficient for selecting the optimal weights is represented by Eq. (5).

$$OB_j = \arg \min_{\{KL_w^{Op}\}} \left(\frac{1}{Ceff} \right) \quad (5)$$

Here, the term KL_w^{Op} indicates optimized weight, and it is taken in the range of [0.01, 0.99]. Weight is the vector that sizes the feature count F_w^{FT} .

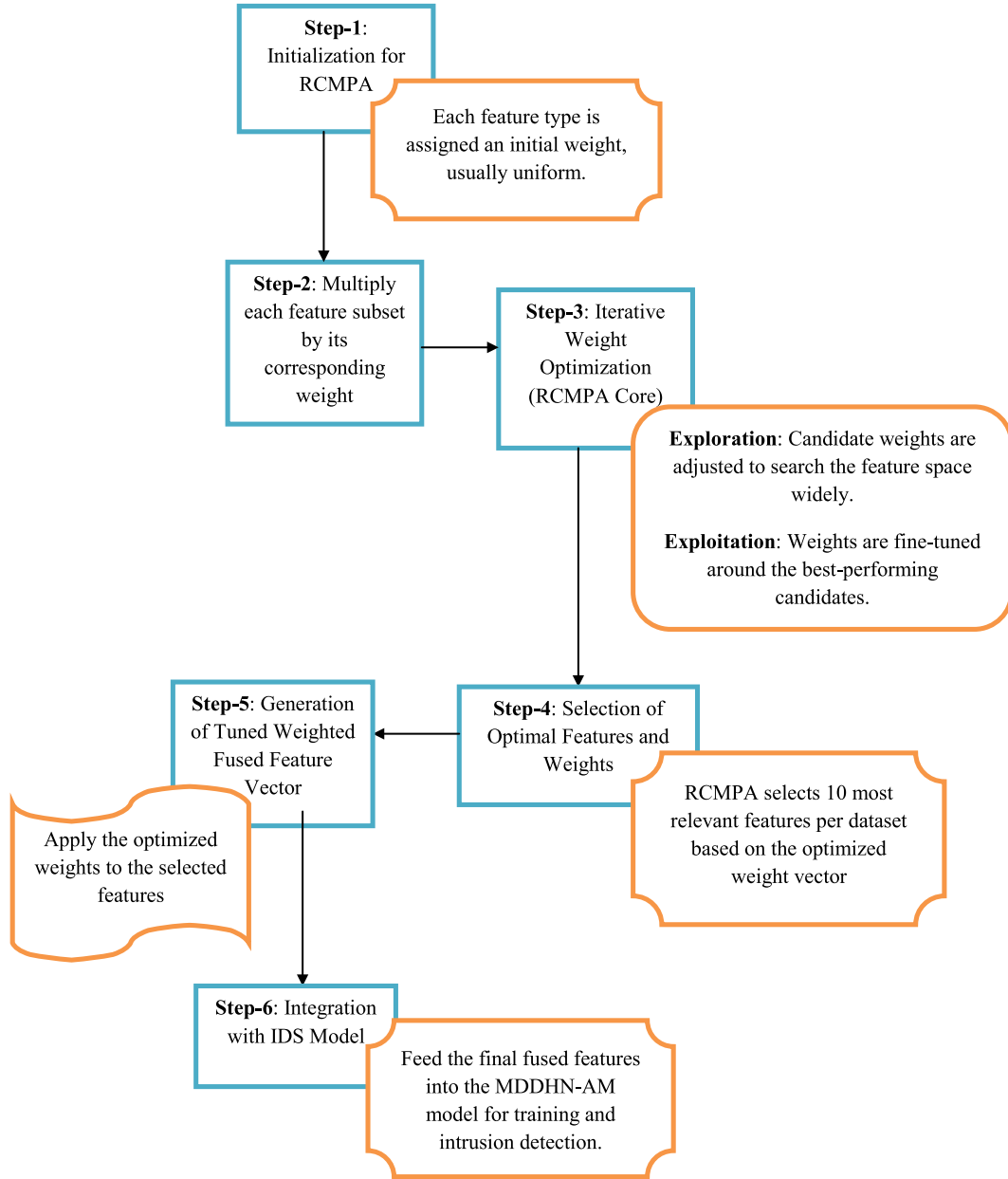


Fig. 6. Steps of RCMPA in Tuned Weighted Feature Fusion.

The correlation coefficient value is measured using Eq. (6).

$$Ceff = \frac{\sum (dk - \bar{d})(ok - \bar{o})}{\sqrt{\sum (dk - \bar{d})^2 \sum (ok - \bar{o})^2}} \quad (6)$$

Here, the value $\bar{d}$ is the mean parameter. The sample parameters are signified as ok and dk , respectively. The final tuned weighted feature is noted by F_w^{FT} . A structural design of a tuned weighted characteristic fusion is shown in Fig. 5. The steps of RCMPA in tuned weighted feature fusion is given in Fig. 6.

5. Developed deep hybrid network with attention mechanism for detecting intrusions to provide higher security

5.1. Basic TCNN

The TCNN [63] model contains two constraints. TCN includes causal

convolution to ensure time and data from the previous time frame to repeat into the process. The normal convolution layer is represented by Eq. (7).

$$(G \oplus Y)(y_k) = \sum_{j=1}^L g_j y_{k-L+j} \quad (7)$$

Here, the value L represents the kernel size. The time is indicated by u .

Dilated convolution: Dilated convolution includes filters that are applied to expand the layer, and this step of expansion helps in capturing a wider range of feature details from data. The expanded convolution in the TCNN is noted by Eq. (8).

$$(G \oplus_e Y)(y_k) = \sum_{j=1}^L g_j y_{k-(L-j)e} \quad (8)$$

Here, the expansion parameter is noted by e .

Residual connection: Deep network training is made simpler by the

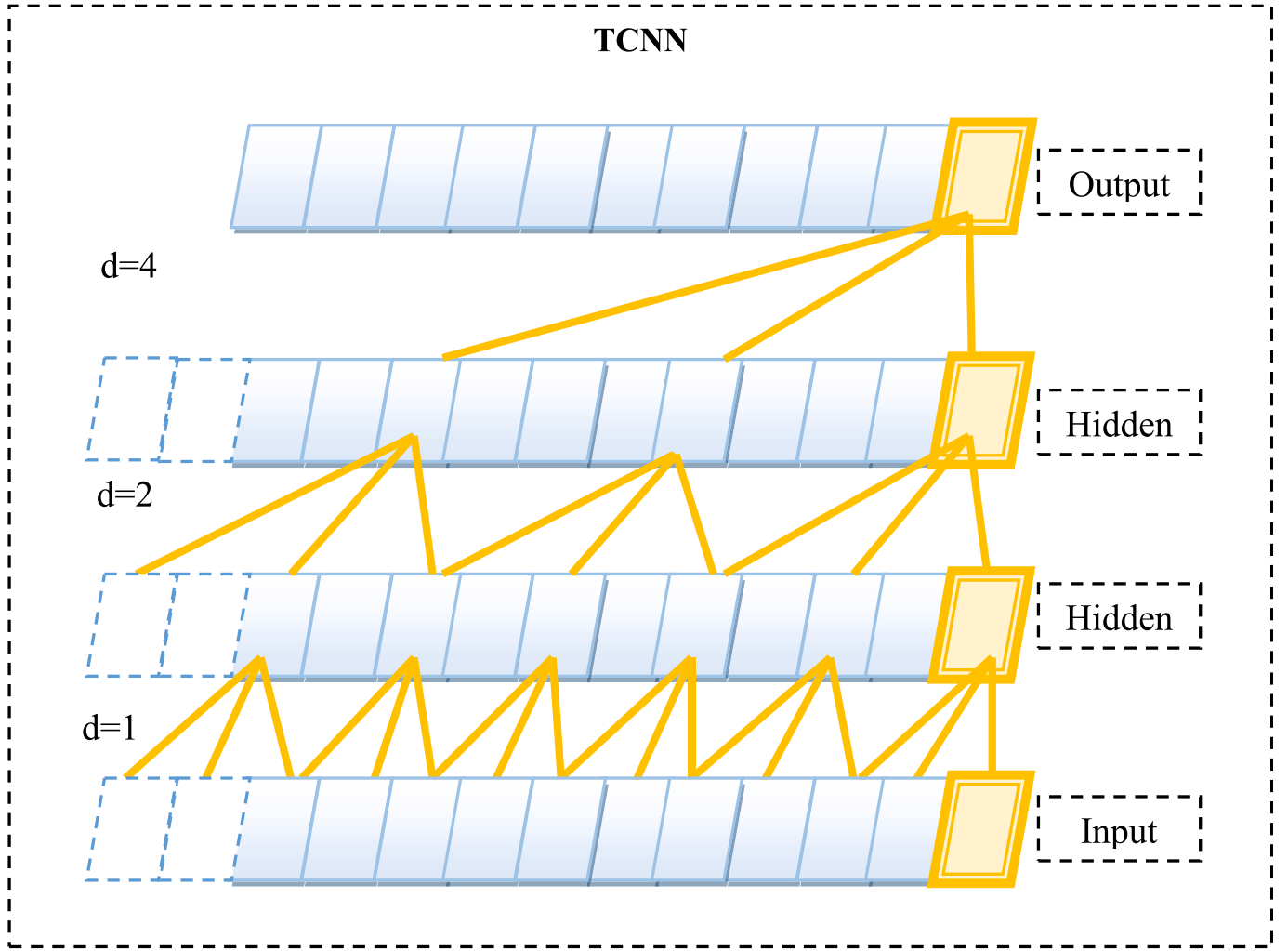


Fig. 7. Basic representation of TCNN.

residual connection. It has been demonstrated that the deep network infused with residual connections is quite efficient and can speed up training while preventing gradient vanishing. The basic design of the TCNN is depicted in Fig. 7.

5.2. Basic RNN

The RNN [64] method is the traditional neural network based on recurrent learning. The Feed-Forward Network (FFN) is part of a RNN process. A feed-forward loops serve as the short-term memory for storing and retrieving historical data over time scales and carry out temporal tasks. To prevent overfitting while training an arbitrary sequence from unknown information, the RNN offers shared values. The RNN's computational process is expressed in Eq. (9).

$$I(y, i) = g(x_{yi}^v + x_{ii}^{i+c}) \quad (9)$$

Here, the term x_{ii}^{i+c} is the weight measure. The output and input are noted by pu_u and y , respectively. The value β is the sigmoid function, and a hidden level is noted by g . The output pu_u is validated by Eq. (10).

$$pu_u = tg(x_{puu}i_u + c_{pu}) \quad (10)$$

Here, the term tg is the soft-max layer. The RNN's loss function is measured by Eq. (11).

$$M = e(us, qs) = \sum_{j=1}^u e(us, qs) \quad (11)$$

Here, the term us is the time series and qs denotes the detected probability distribution. The RNN method contains weight parameters. The RNN's cyclic connections identify the output by rearranging the input sequences from different connectivity and thus allow for better gradient flow. The basic design of the RNN method is depicted in Fig. 8.

5.3. Proposed MDDHN-AM-based ID

Deep learning techniques, when hybridized have increased the capability to manage huge information and learn a features effectively. Here, a hybridized network is developed, inspired by RNN and TCNN, and an attention mechanism is used to engage in intrusion detection. The final tuned fused weighted feature from the feature extraction phase F_w^{FT} is fed into the MDDHN-AM network for intrusion detection. The TCNN and RNN method is hybridized through an attention mechanism to develop the MDDHN-AM strategy.

Reason for Selecting the Models: The selection of RNNs and TCNNs for ID is based on their unique capabilities to handle time series and sequential data, which are essential in dealing with ID datasets. The ID datasets contain a lot of sequential information, making recurrent learning highly suitable. RNNs are ideal for processing sequential data because they maintain a memory of previous inputs, allowing them to

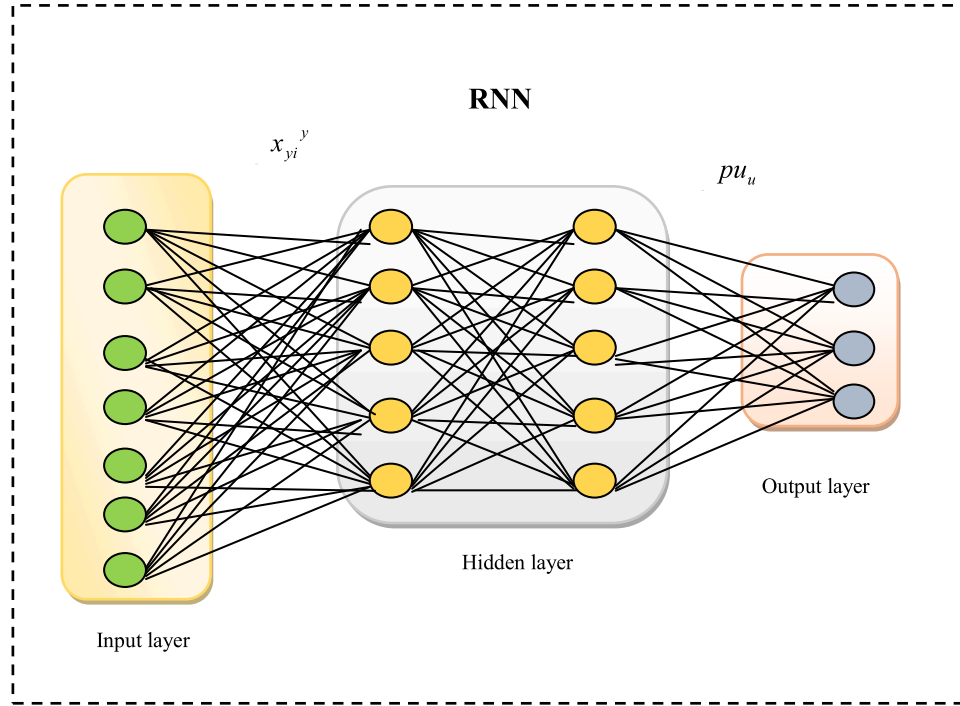


Fig. 8. Diagrammatic representation of RNN.

identify patterns and dependencies over time. This is important for diagnosing attacks that unfold gradually, such as DoS attacks, where the intrusion occurs over a sequence of actions rather than a single event. However, traditional RNNs can be computationally intensive, especially with long sequences. To overcome this, TCNNs are employed, as they use convolutional operations with dilated and residual connections and efficiently capture local temporal patterns, allowing them to process data in parallel. This results in faster computations and the ability to handle large volumes of network traffic more efficiently. TCNNs also utilize dilated convolutions, enabling them to learn long-range temporal dependencies without increasing the number of parameters, making them well-suited for detecting short-term as well as long-term attack patterns. By combining the strengths of both models, the hybrid MDDHN-AM methodology gives an efficient and vigorous solution for recognizing complex intrusion behaviors in network traffic.

Novelty: The MDDHN-AM introduces a new methodology to ID by leveraging a synergistic strength of different deep learning models. The inputs are given in the format of multi-scale dilated convolution in the MDDHN-AM method. The use of multi-scale dilated convolutions permits the network to detain large-range dependencies across different time intervals, which is crucial for detecting both immediate and gradual changes in network behavior. Traditional models may struggle to identify evolving attacks; however, the multi-scale approach ensures that MDDHN-AM can detect a wider range of attack patterns, from rapid intrusions to slow and stealthy threats. This makes the model highly effective for real-world scenarios, which are not always predictable.

Dilated convolution is the process of enlarging the convolution kernel by inserting some extra spaces between the convolution kernel's constituent parts. The feature inputs are given in the format of multi-scale dilated convolution in the MDDHN-AM method. The input is noted by $\{y_1, y_2, \dots, y_u\}$. The kernel size is calculated using Eq. (12).

$$L = L + (l - 1)(s - 1) \quad (12)$$

Here, the term l is the original size of the Convolutional kernel. The size is indicated by L . The dilation rate is specified by s . It is used to calculate the expansion coefficient. Dilated convolution has the drawback of the grid effect due to overstacking, yet it can large a interested

field and raise a resolution of an outputs. The actual convolution size is 3×3 . The term s is set to $s = 1, 2, 4$. The term $\{g_1, g_2, \dots, g_u\}$ is the hidden layer. The weight is noted by $\{\beta_1, \beta_2, \dots, \beta_u\}$. The present weight is specified by $\{f_1, f_2, \dots, f_u\}$. Final convolution level's output is set to the size of 1×1 . The multi-scale information is given in the form of 4 dilation rates.

Then, the obtained output is given to the hybrid TCNN and RNN models. The hybridization of TCNNs and RNNs in MDDHN-AM enables the model to handle both sequential and time-series data, which is essential for accurate ID. TCNNs are suitable for detaining local characteristics and temporal dependencies in the information, while RNNs are designed to process sequences and learn long-term dependencies, making the combination ideal for modeling network traffic and identifying abnormal behavior. To enhance performance, the attention mechanism is used to focus on a more appropriate characteristics, enhancing both computational efficiency and accuracy. It permits a model to assign more significance to important data points, such as suspicious traffic patterns or sudden deviations, while ignoring less relevant information. This attention-driven approach helps in reducing false positives and ensures that the model prioritizes critical events, leading to more reliable and targeted intrusion detection.

The attention mechanism contains both spatial and channel attention. Inserting attention in the developed MDDHN-AM method decreases the complexity of calculation. The weighting procedure of the attention mechanism is given in Eq. (13).

$$P = B(m)^\circ m \quad (13)$$

Here, the term $B(m)$ is bias. The element-wise product is noted by $^\circ$. The output is indicated by $\{z_1, z_2, \dots, z_u\}$. The developed MDDHN-AM method is utilized to predict the higher-order information and identify the best features. At last, the proposed MDDHN-AM method offers the ID-detected result.

Advantages of Developed MDDHN-AM: The MDDHN-AM model stands out due to its adaptability to different types of attacks, including those with varying attack patterns and timeframes. By effectively integrating multiple layers of feature extraction, temporal analysis, and attention mechanisms, the developed MDDHN-AM can identify subtle

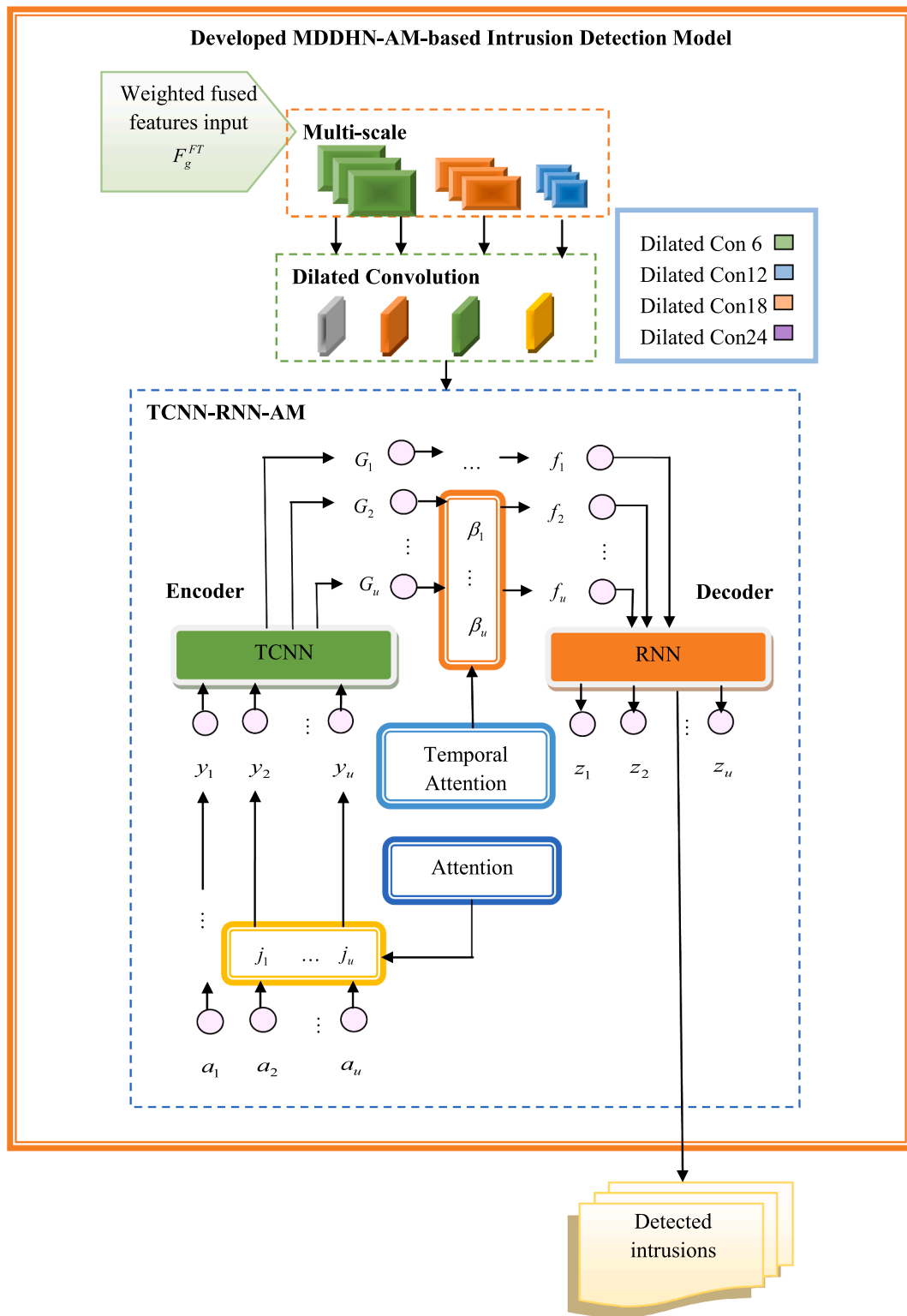


Fig. 9. Structural design of the investigated MDDHN-AM-based ID System.

anomalies and sophisticated intrusions that may otherwise go undetected in traditional systems. Finally, the MDDHN-AM is a novel and highly effective intrusion detection model that combines multi-scale dilated convolutions, hybridized temporal networks, and an attention mechanism to deliver superior performance in detection. Its ability to adjust to multipart attack patterns, prioritize critical data, and effectively learn from temporal dependencies makes it a powerful tool in safeguarding modern network infrastructures. The structural design of

the implemented MDDHN-AM-based ID is depicted in Fig. 9.

5.4. Differences from existing optimization + hybrid deep learning IDS models

Most of the conventional systems used for IDS combine the power of a deep learning model with optimization algorithm, thus enhancing the performance of detection. But these approaches focus on using

Table 2
Configuration table.

Category	Parameter	TCNN	RNN
Input Architecture	Input shape	1520, 30	1520, 30
	Total layers	5	3
	Size of Filter	3	–
	Dilation rate of the convolution layer	2 (Exponential Increase)	–
	Strides	1 or 2	–
	Hidden neurons in each layer	32	256
Recurrent	Activation	ReLU	Tanh
	Cell type	–	LSTM
	Dropout rate	0.1	0.5
Regularization	Batch	Yes	No
	Weight	L1/L2 0.01	L1/L2 0.1
Training	Loss	Cross Entropy	Cross Entropy
	Optimizer	RMSprop	Adam
	Learning rate	0.001	0.001
	Batch size	16	64
	Epoch	100	200
	Software requirement	pycharam, version 3.11 and anaconda, version 3	
	Hardware Requirement	Machine – Windows Version – 11 Processor – i3 RAM – 8 GB Hard disk space – 500 GB	
Libraries	Keras		
	TensorFlow		
	tflearn		
	matplotlib		
	numpy prettytable		

optimization algorithms only for optimal feature selection or optimizing the detection network's parameters, lacking the simultaneous performance combination. For example, the existing model, named IntrusionX [65], was developed by combining the Convolutional-LSTM with the Squirrel Search Optimization (SSO) for ID. Another study used HHO-MLP [66], developed using the power of Harris Hawks Optimization and Multi-Layer Perceptron. However, these models used optimization algorithms, such as SSO and HHO, for fine-tuning the model parameters. Although these models are effective, they are not effective in addressing the feature fusion optimization and integration with the deep learning model concurrently. Unlike these, the developed model introduces an innovative strategy, which has synergy between the optimization algorithm (RCMPA) and the deep learning model (MDDHN-AM). Here, the RCMPA is used for tuning the weighted feature fusion phase, which ensures that only the essential features are taken throughout the process of learning. The existing models overlooked addressing the imbalance and feature redundancy issues, but the developed model addresses these problems by using the RCMPA. Moreover, MDDHN-AM also proves its differentiation from the existing models as it integrates the strengths of TCNN and RNN, along with an attention mechanism. To provide enhanced ability in detecting the complicated and unknown assault patterns, this combination is essential as these allows for modeling spatial and temporal dependencies simultaneously. Thus, by concurrently optimizing feature fusion and deep learning model integration, the developed methodology provides the most comprehensive and adaptive solution for intrusion diagnosis, setting it apart from existing optimization-based hybrid deep learning approaches.

6. Results and discussions

6.1. Experimental setup

“A proposed ID network was developed using Python software, leveraging several deep learning libraries, including TensorFlow and

Keras for model development, and Scikit-learn for data preprocessing and evaluation. The primary objective was to design, optimize, and evaluate an ID system capable of detecting network intrusions with high accuracy and efficiency. For model optimization and training, several hyperparameters were tuned. The population size of the genetic algorithm used in the RCMPA optimization was set to 10. The chromosomal distance size, representing the number of genes in each individual of the population was set to 4, and the maximum number of iterations was set to 50.” These settings were chosen based on a balance between system complexity as well as computational efficiency, ensuring an effective search for the optimal feature weights while keeping training time reasonable. The heuristic approaches, including CMBA [59], Osprey Algorithm (OOA) [40], Henry Gas Solubility (HGSO) [68], and Tomtit Flock Metaheuristic Optimization Algorithm (TFMOA) [69] and the traditional techniques, including Deep Belief Networks (DBN) [33], TCNN [63], RNN [64] and TCNN-RNN [63 64] were employed for the testing investigation. In addition to the binary class attack setting, a multi-class attack analysis was carried out on a wider range of datasets to ensure the system's adaptability to new attack classes and generalization ability. Table 2 gives the parameter setup for training the developed ID model.

6.2. Evaluation measures

The performance of the developed ID system was evaluated using multiple metrics such as accuracy, precision, F1-score, and others, compared with existing strategies. Each metric provides valuable insights into the system's effectiveness in detecting abnormal activities and ensuring network security. By evaluating the developed ID system across these metrics, a comprehensive assessment of its performance is achieved, allowing for a thorough comparison with other techniques. Each metric contributes to understanding different aspects of the system's detection capabilities and reliability, helping to identify areas of improvement and optimize its performance.

(a) Accuracy: Accuracy is a measure of the overall system's correctness, evaluating the ratio of correctly classified attacks and normal traffic (both positive and negative) to the total instances. While accuracy is a basic metric, it is useful when the dataset is balanced and all classes are of equal importance.

$$acy = \frac{(ZU_c + OR_i)}{(ZU_c + OR_i + ZU_d + OR_j)} \quad (14)$$

(b) False Negative Rate (FNR): FNR quantifies the positive instances (real attacks) that are not correctly identified as negative. Minimizing FNR ensures that potential threats are not overlooked by the ID system.

$$fnr = \frac{OR_i}{OR_i + ZU_c} \quad (15)$$

(c) False Positive Rate (FPR): FPR measures the negative instances that are not correctly classified as positive. A lower FPR indicates that the system is effective in minimizing false alarms and is crucial for maintaining system trust.

$$fpr = \frac{ZU_d}{ZU_d + OR_i} \quad (16)$$

(d) False Discovery Rate (FDR): FDR measures the proportion of false positives among all instances identified as positive. Lower FDR values indicate that the system is better at accurately detecting intrusions without unnecessary false alarms.

$$fdr = \frac{OR_i}{ZU_d + OR_j} \quad (17)$$

(e) F1-score: The F1-score is the harmonic mean between precision and sensitivity. This metric balances the trade-off between false positives and false negatives, providing a comprehensive evaluation of the

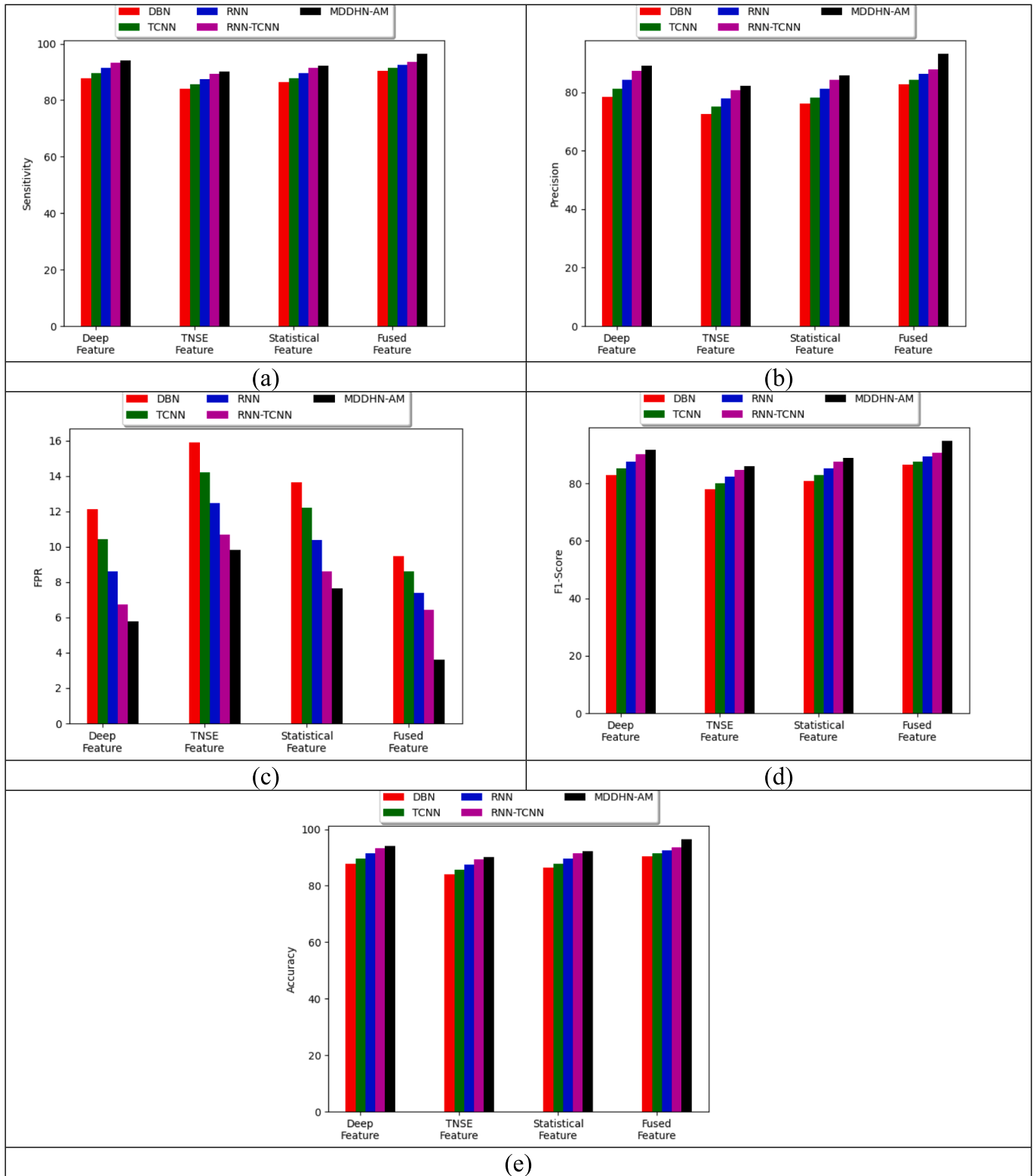


Fig. 10. Evaluation of the proposed ID system against Deep learners on Dataset 1 with (a) Sensitivity, (b) Precision, (c) FPR, (d) F1-score and (e) Accuracy.

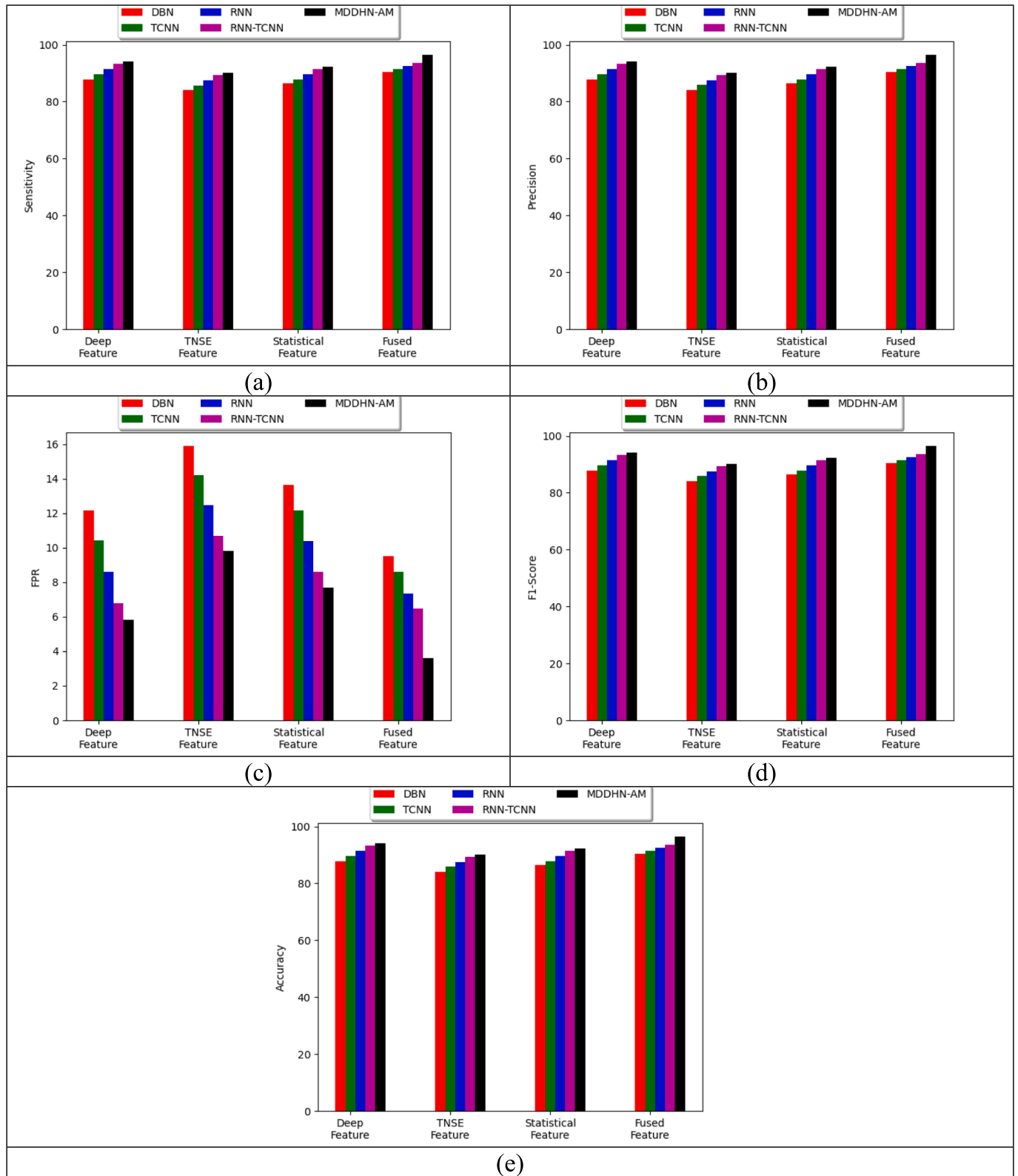


Fig. 11. Evaluation of the suggested ID system Against Different Deep learners on Dataset 2 with (a) Sensitivity, (b) Precision, (c) FPR, (d) F1-score and (e) Accuracy.

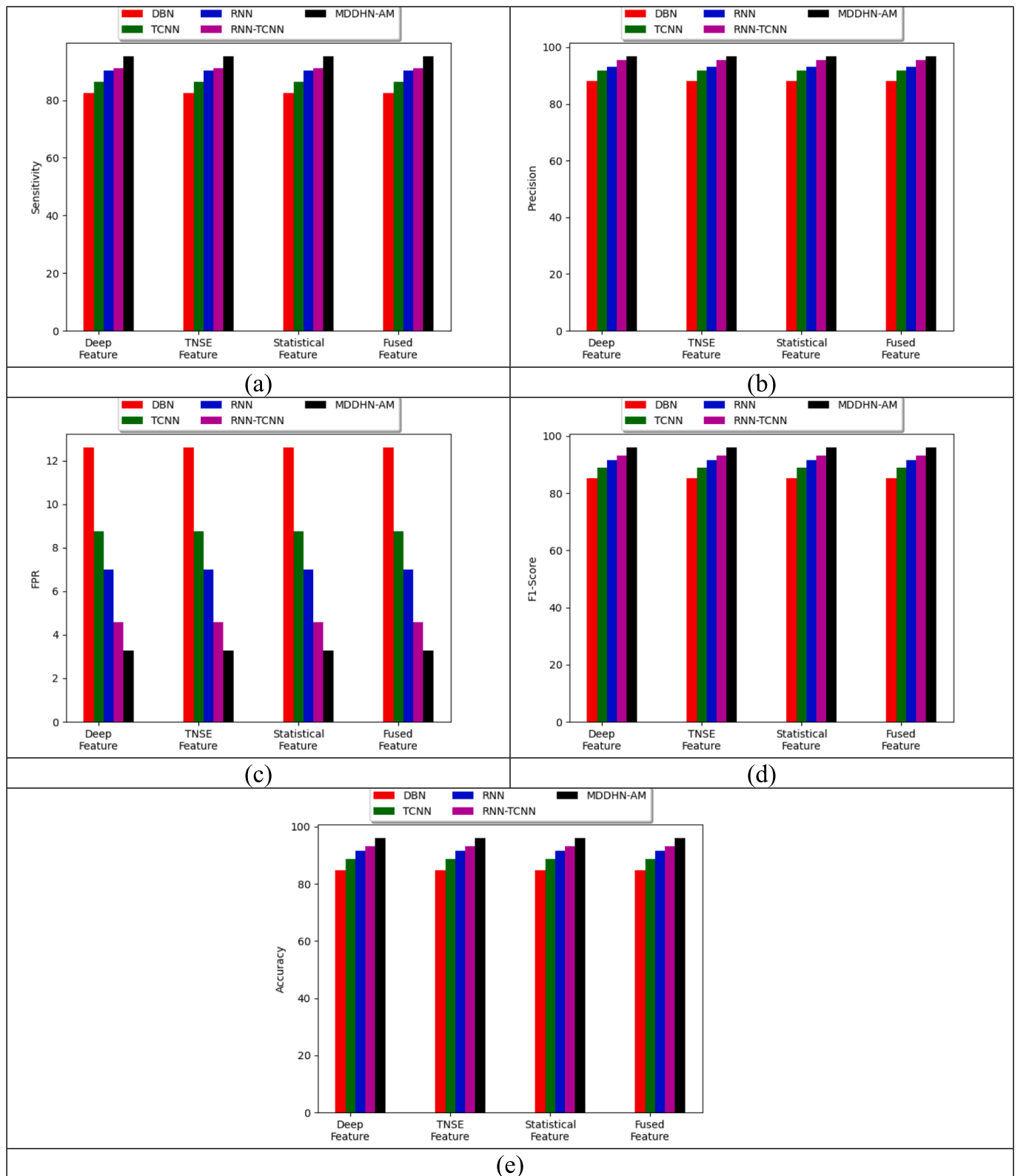


Fig. 12. Evaluation of the developed ID system Against Different Deep learners on Dataset 3 with (a) Sensitivity, (b) Precision, (c) FPR, (d) F1-score and (e) Accuracy.

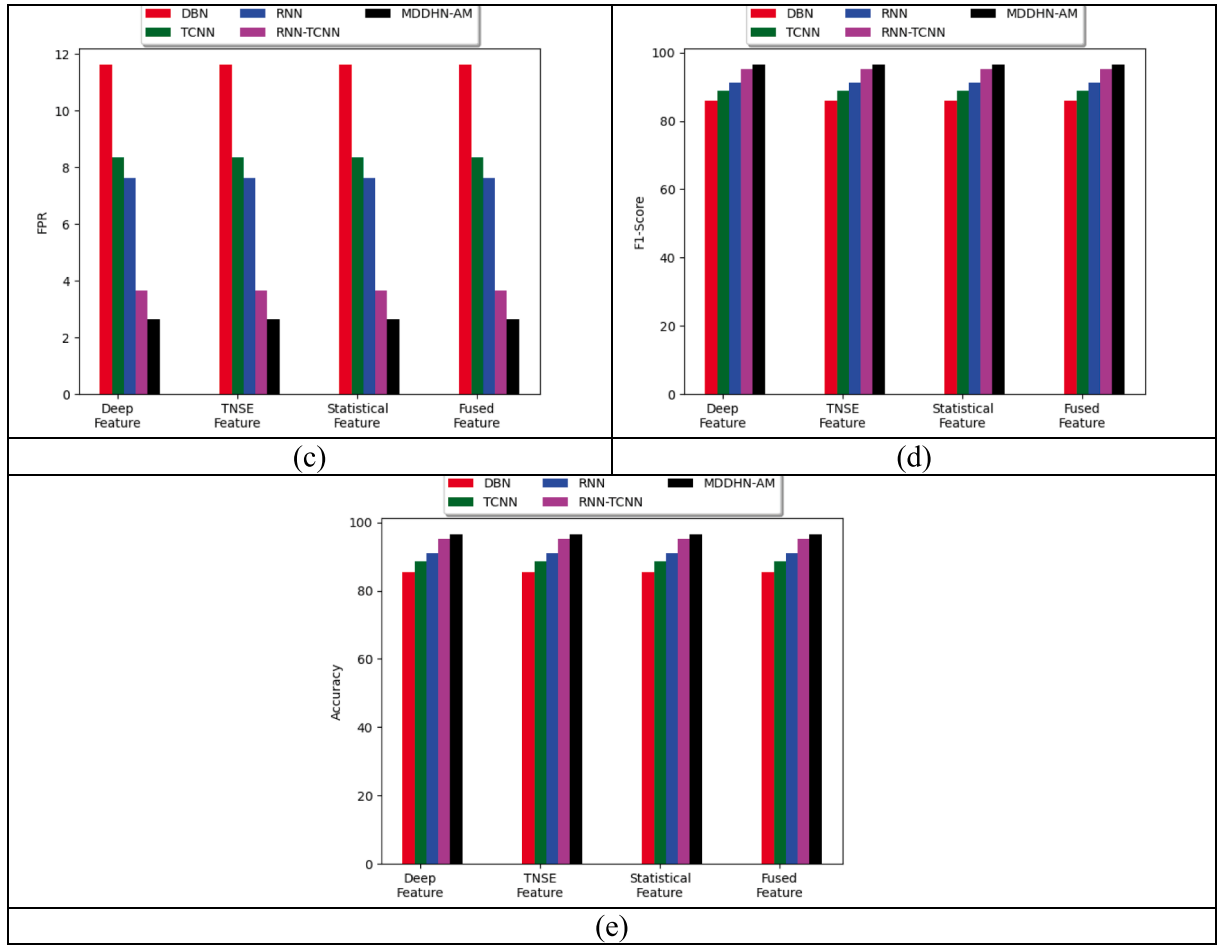


Fig. 13. Evaluation of the developed ID system Against Different Deep learners on Dataset 4 with (a) Sensitivity, (b) Precision, (c) FPR, (d) F1-score and (e) Accuracy”.

system’s ability to detect intrusions while minimizing errors. It is particularly useful for the dataset under the uneven distribution of classes.

$$F1 = \frac{2 \times ZU_c}{2ZU_c + OR_i + OR_j} \quad (18)$$

(f) Specificity: This metric measures the proportion of actual negative instances that are correctly identified. It is essential to evaluate how well the system avoids false positives, ensuring that normal network activities are not flagged as intrusions.

$$SPE = \frac{OR_i}{OR_i + ZU_d} \quad (19)$$

(g) Sensitivity or Recall: Sensitivity, also known as recall or true positive rate, assesses the ability of the ID system to correctly identify positive instances, i.e., actual intrusions. It is crucial for ensuring that the system does not miss potential threats and provides a high detection rate for real attacks.

$$SEN = \frac{ZU_c}{ZU_c + OR_i} \quad (20)$$

(h) Precision: Precision evaluates the proportion of true positive results out of all instances identified as positive by the system. A higher precision ensures that the system is not overwhelmed by false alarms, thus enhancing its reliability in predicting actual intrusions.

$$PRE = \frac{OR_i}{OR_j + ZU_d} \quad (21)$$

(i) NPV: NPV measures the proportion of correctly predicted negative cases (non-intrusions) among all instances classified as negative by the system. It is important to ensure that the system does not incorrectly classify benign activities as non-intrusions, thereby minimizing false negatives.

$$NPV = \frac{ZU_d}{ZU_d + OR_j} \quad (22)$$

(j) Matthews correlation coefficient (MCC): MCC provides a comprehensive evaluation of the system’s performance by considering true positives, true negatives, false positives, and false negatives in a single value. It is particularly useful for imbalanced datasets, where traditional metrics like accuracy may be misleading.

$$MCC = \frac{ZU_c \times ZU_d - OR_j \times OR_i}{\sqrt{(ZU_c + OR_j)(ZU_c + OR_i)(ZU_d + OR_j)(ZU_d + OR_i)}} \quad (23)$$

Here, the parameter ZU_c is the true positive and the parameter ZU_d is the true negative. The variables OR_i and OR_j are the false negatives and positives, respectively.”.

6.3. Presentation evaluation on developed model using dataset-1

Fig. 10 shows an effectiveness of a designed ID system compared to various detection approaches under Dataset 1. The MDDHN-AM-based ID system outperformed other techniques with a high precision of 5.79 % to DBN, 4.05 % to TCNN, 8.57 % to RNN, and 10.59 % to TCNN-RNN, respectively. The network’s ability to process varied feature types,

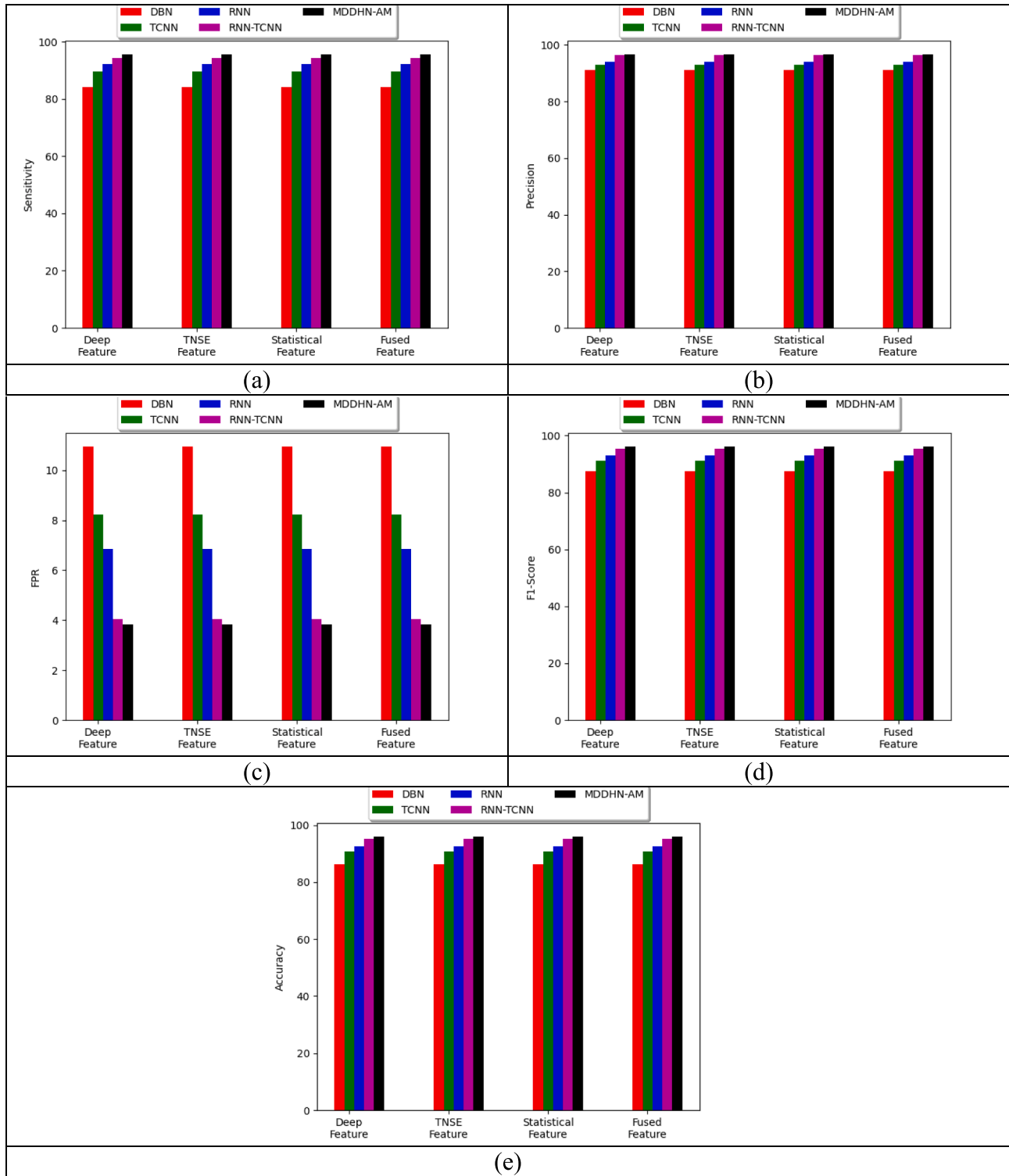


Fig. 14. Evaluation of the developed ID system Against Different Deep learners on Dataset 5 with (a) Sensitivity, (b) Precision, (c) FPR, (d) F1-score and (e) Accuracy”.

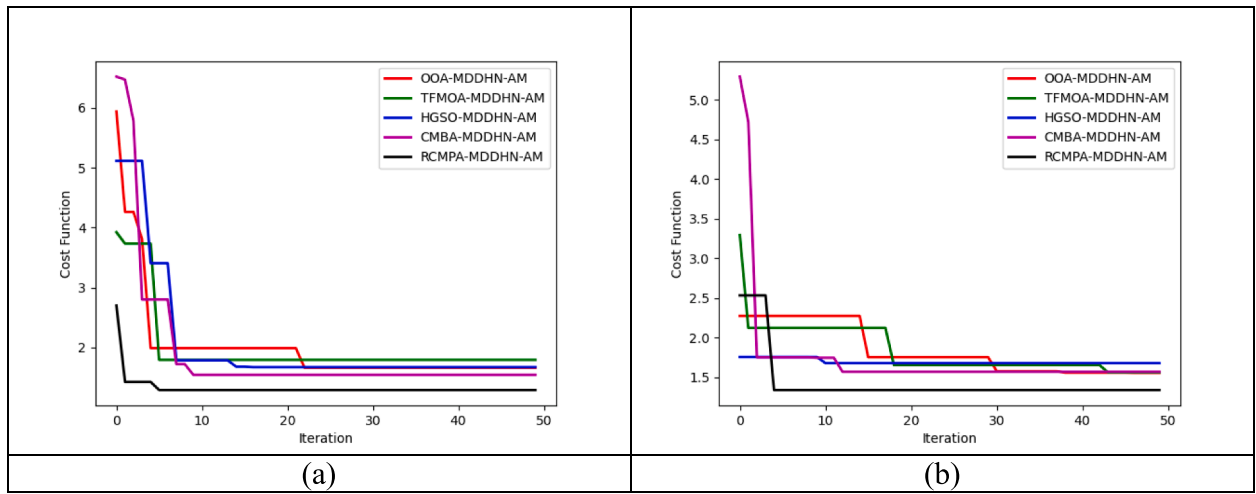


Fig. 15. Cost function testing on developed ID system among various algorithms concerning (a) dataset-1 and (b) dataset-2^o.

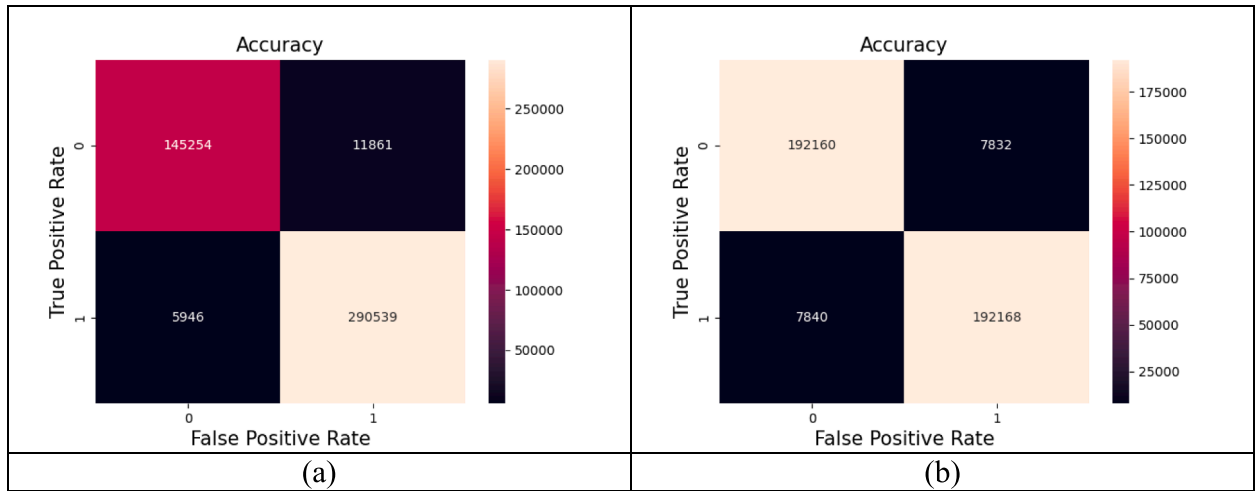


Fig. 16. Confusion matrix of the developed ID system among several methods and algorithms with respect to (a) dataset-1 and (b) dataset-2^o.

like statistical, temporal, and deep learning-based features, ensures a comprehensive analysis of network traffic, makes it a more reliable solution for latest network security limitations. As a result, the MDDHN-AM-based ID system is positioned as a more effective and efficient solution for ID in complex and dynamic network environments.

6.4. Performance analysis on the offered system using dataset-2

Fig. 11 presents the comparative study of the developed MDDHN-AM against deep learners on Dataset 2. The MDDHN-AM-based ID system achieved a higher f1-score of 5.79 % than DBN, 4.05 % than TCNN, 8.57 % than RNN, and 10.59 % than TCNN-RNN while using the fused feature set. Also, this analysis indicates the advantage of using the fused feature set in training, as it improved the performance of all the comparative models. Also, the proposed model has ensured for high precision rate of 96 % showing its effectiveness in processing different attacks. The MDDHN-AM-based ID system provided better efficacy than other frameworks with a high F1-score. Thus, it proved that the developed MDDHN-AM-based ID system demonstrated superior performance compared to several other detection models. The system effectively combined deep learning techniques, feature extraction, and optimized hybrid models to achieve a robust and accurate detection capability. Moreover, its ability to outperform traditional and existing models showcases its effectiveness in addressing the challenges associated with

intrusion detection tasks, making it a promising approach for enhancing network security.”.

6.5. Presentation analysis of proposed MDDHN-AM on Dataset 3, Dataset 4 and Dataset 5

Fig. 12, Fig. 13, and Fig. 14 show the comparative description of developed MDDHN-AM model under Dataset 1, Dataset 2, and Dataset 3, respectively. Here, as indicated through the comparative graphs, the presentation of a MDDHN-AM is comparatively greater than other existing models, considering different performance measures. On Dataset 3, the MDDHN-AM has achieved values for Accuracy, Sensitivity, Precision, F1-score, and FPR as 95.9 %, 95.1 %, 96.7 %, 95.9 %, and 3.27 % respectively for while using the fused feature set. Similarly, for Dataset 4, the MDDHN-AM has achieved values for Accuracy, Sensitivity, Precision, F1-score, and FPR as 96.4 %, 95.5 %, 97.3 %, 96.4 %, and 2.6 % respectively, while using the fused feature set. Finally, for Dataset 5, the MDDHN-AM has attained values for Accuracy, Sensitivity, Precision, F1-score, and FPR as 95.8 %, 95.5 %, 96.6 %, 96.09 %, and 3.8 %, respectively, while using the fused feature set. This brief analysis indicates that the proposed MDDHN-AM performs well under different intrusion datasets, and the ability of the feature selection has a great impact on categorizing the intrusion class. The inclusion of the attention mechanism and multi-scale dilated networks further enhanced its

Table 3

Performance estimation of developed ID system model among different algorithms.

Terms	OOA-MDDHN-AM [67]	TFMOA-MDDHN-AM [69]	HGSO-MDDHN-AM [68]	CMBA-MDDHN-AM [59]	RCMPA-MDDHN-AM
Dataset-1					
Accuracy	90.48611	91.39153	93.23236	94.64793	96.07429
Sensitivity	90.47553	91.39947	93.23148	94.64815	96.06746
Specificity	90.4914	91.38757	93.2328	94.64782	96.07771
Precision	82.63154	84.14272	87.3233	89.83948	92.45075
FPR	9.508598	8.612434	6.767196	5.352183	3.922288
FNR	9.524471	8.600529	6.768519	5.351852	3.93254
NPV	95.00047	95.50594	96.49724	97.25049	97.9945
FDR	17.36846	15.85728	12.6767	10.16052	7.549247
F1-Score	86.37582	87.6211	90.18072	92.18114	94.22441
MCC	0.792819	0.812027	0.851322	0.881861	0.912913
Dataset-2					
Accuracy	90.49375	91.40825	93.24025	94.65525	96.082
Sensitivity	90.4965	91.4015	93.232	94.6625	96.08
Specificity	90.491	91.415	93.2485	94.648	96.084
Precision	90.49152	91.41384	93.24739	94.64878	96.08384
FPR	9.509	8.585	6.7515	5.352	3.916
FNR	9.5035	8.5985	6.768	5.3375	3.92
NPV	90.49598	91.40266	93.23312	94.66173	96.08016
FDR	9.508477	8.586159	6.752614	5.351224	3.916157
F1-Score	90.49401	91.40767	93.23969	94.65564	96.08192
MCC	0.809875	0.828165	0.864805	0.893105	0.92164

Table 4

Performance estimation of the developed ID system among different Methods.

Terms	DBN [33]	TCNN [63]	RNN [64]	RNN-TCNN [63 64]	MDDHN-AM
Dataset-1					
Accuracy	89.12765	90.03197	91.38845	92.778	96.07429
Sensitivity	89.11177	90.04167	91.37765	92.76984	96.06746
Specificity	89.13558	90.02712	91.39385	92.78208	96.07771
Precision	80.39632	81.86541	84.14927	86.53444	92.45075
FPR	10.86442	9.972884	8.606151	7.217923	3.922288
FNR	10.88823	9.958333	8.622354	7.230159	3.93254
NPV	94.24389	94.75912	95.49535	96.24981	97.9945
FDR	19.60368	18.13459	15.85073	13.46556	7.549247
F1-Score	84.52999	85.7591	87.61462	89.54372	94.22441
MCC	0.764225	0.783277	0.81193	0.841567	0.912913
Dataset-2					
Accuracy	89.14175	90.034	91.3845	92.79125	96.082
Sensitivity	89.1495	90.021	91.3775	92.796	96.08
Specificity	89.134	90.047	91.3915	92.7865	96.084
Precision	89.13568	90.04441	91.39029	92.78719	96.08384
FPR	10.866	9.953	8.6085	7.2135	3.916
FNR	10.8505	9.979	8.6225	7.204	3.92
NPV	89.14782	90.02359	91.37871	92.79532	96.08016
FDR	10.86432	9.955588	8.609705	7.212815	3.916157
F1-Score	89.14259	90.0327	91.3839	92.79159	96.08192
MCC	0.782835	0.80068	0.82769	0.855825	0.92164

capability to detain intricate patterns and anomalies in system traffic. This allows the classifier to be used under different complex conditions and detect different attack types.

6.6. Cost function examination of over-developed algorithm

In this stage, the convergence performance over other heuristic algorithms and an outcome are shown in Fig. 15. When employing dataset 1 at an iteration value of 20, the RCMPA-MDDHN-AM-based ID system achieved a decreased cost function of 5.79 % to OOA-MDDHN-AM, 4.05 % to TFMOA-MDDHN-AM, 8.57 % to HGSO-MDDHN-AM, and 10.59 % to CMBA-MDDHN-AM. The RCMPA-MDDHN-AM-based ID provided a

low-cost function when compared to existing frameworks. Thus, it proved that the developed RCMPA-MDDHN-AM system demonstrated superior convergence performance compared to other heuristic algorithms. As the cost function was defined based on maximizing the correlation coefficient, the low cost function indicates a better correlation among the fused features through the newly developed RCMPA algorithm. This highlights the potential of the RCMPA-MDDHN-AM-based ID system as a reliable solution, offering improved performance in detecting anomalies and minimizing detection costs.

6.7. Confusion matrix evaluation on the developed framework

Fig. 16 illustrates an effectiveness of the developed ID systems through the confusion matrix analysis for a binary attack class setting. A presentation of a RCMPA-MDDHN-AM-based ID demonstrated a greater accuracy of 95.75 % in detecting the traffic as normal or under attack. The high accuracy demonstrated by the RCMPA-MDDHN-AM-based model shows its effectiveness in correctly classifying intrusion events. The outcomes evaluate an efficiency of combining the RCMPA optimization technique with the MDDHN-AM model, which provides a important enhancement in terms of diagnosis accurateness and an overall system presentation.

6.8. Effectiveness testing over heuristic algorithms

The RCMPA-MDDHN-AM-based ID structure showed better accuracy of 96.07 % which is higher than the OOA-MDDHN-AM, TFMOA-MDDHN-AM, HGSO-MDDHN-AM, and CMBA-MDDHN-AM for dataset-1. Thus, it proved that the RCMPA-MDDHN-AM-based ID system demonstrates its superior accuracy compared to other heuristic algorithms, as shown in Table 3. By leveraging the optimal weighted fused features, the RCMPA-MDDHN-AM-based ID system exhibited a higher level of performance in detecting intrusions. The outcomes underscores an efficiency of the developed ID system in providing more reliable intrusion detection, outperforming traditional methods, and offering a promising approach for network security.

6.9. Effectiveness analysis over traditional methods

The RCMPA-MDDHN-AM-based ID outcomes achieved an improved accuracy of 96.02 %, which is higher than the DBN, TCNN, RNN, and TCNN-RNN on dataset 2, as indicated in Table 4. In addition, the F1-score, precision, and sensitivity of the developed approach are higher than the existing techniques, showcasing the system's ability to solve the class imbalance issues. The model better handled the data imbalance issues and achieved a high F1-score of 94.2 % under Dataset 1. Thus, it proved that the developed model demonstrated superior performance over traditional detection methods. By utilizing the extracted features and optimizing them through the RCMPA, the system was able to achieve enhanced accuracy, precision, and sensitivity. The results suggest that the proposed RCMPA-MDDHN-AM-based ID system outperforms conventional methods, making it a robust solution for detecting anomalies in network traffic and contributing to better network security.”.

6.10. Statistical evaluation among heuristic strategies

A statistical evaluation of the proposed ID framework is displayed in Table 4. The RCMPA-MDDHN-AM-based ID structure provided an enhanced best value of 50 % than OOA-MDDHN-AM, 52.68 % over TFMOA-MDDHN-AM, 52.47 % than HGSO-MDDHN-AM, and 46.77 % than CMBA-MDDHN-AM using dataset-1. The developed ID model provided better efficacy than another scheme. Thus, it proved that the statistical analysis of the RCMPA-MDDHN-AM-based ID framework demonstrated significant improvements over other heuristic approaches, as evidenced by the results in Table 5. By incorporating optimized feature fusion and utilizing advanced deep learning

Table 5

Statistical evaluation of recommended ID model among various algorithms.

Terms	OOA-MDDHN-AM [67]	TFMOA-MDDHN-AM [69]	HGSO-MDDHN-AM [68]	CMBA-MDDHN-AM [59]	RCMPA-MDDHN-AM
Dataset-1					
Best	0.930351	0.886214	0.884553	0.995778	1.865098
Worst	3.794138	2.938147	2.527495	5.985618	1.50421
Mean	1.50529	1.27785	1.276206	1.320873	1.064794
Median	1.281156	1.251954	1.212605	1.168947	1.0447
Standard deviation	0.440511	0.58853	0.513341	0.921169	0.009801
Dataset-2					
Best	1.091086	0.893119	0.865986	1.08748	1.848416
Worst	6.200344	2.267646	3.845125	3.642113	2.019521
Mean	1.391971	1.145949	1.243173	1.673581	1.036489
Median	1.190396	1.273599	1.199844	1.08748	0.093035
Standard deviation	0.834698	0.294835	0.636286	1.004678	0.09871

Table 6

State-of-the-art evaluation with recent ID models.

Terms	DCGAN [39]	GSAFS-OQNN [40]	LSTM-XAI-FA [48]	E-GraphSAGE [45]	MDDHN-AM
Dataset-1					
Accuracy	82.60469	84.7676	86.65439	89.00138	96.07429
Sensitivity	80.7363	83.05964	84.86159	87.42358	96.06746
Specificity	84.77612	86.7126	88.69223	90.75875	96.07771
Precision	86.04015	87.68248	89.5073	91.33212	92.45075
FPR	15.22388	13.2874	11.30777	9.241245	3.922288
FNR	19.2637	16.94036	15.13841	12.57642	3.93254
NPV	79.10864	81.8013	83.75116	86.62953	97.9945
FDR	13.95985	12.31752	10.4927	8.667883	7.549247
F1-Score	83.30389	85.30848	87.12256	89.33512	94.22441
MCC	0.653303	0.696279	0.73406	0.780719	0.912913
Dataset-2					
Accuracy	83.43685	84.33402	87.09455	88.81988	96.082
Sensitivity	81.99234	82.78061	85.78811	87.25869	96.08
Specificity	85.13514	86.16541	88.59259	90.625	96.084
Precision	86.63968	87.58435	89.60864	91.49798	96.08384
FPR	14.86486	13.83459	11.40741	9.375	3.916
FNR	18.00766	17.21939	14.21189	12.74131	3.92
NPV	80.08475	80.9322	84.46328	86.01695	96.08016
FDR	13.36032	12.41565	10.39136	8.502024	3.916157
F1-Score	84.25197	85.11475	87.65677	89.32806	96.08192
MCC	0.669256	0.68731	0.742261	0.776991	0.92164

techniques, the RCMPA-MDDHN-AM-based ID system proved to be a more well-organized and robust solution for intrusion diagnosis, highlighting its superiority in network security tasks compared to traditional models.

6.11. State-of-the-art evaluation of the developed model

A state-of-the-art evaluation of the developed model is given in Table 6. The developed MDDHN-AM model provided the highest accuracy of 96.07 % for dataset-1 and 96.082 % for dataset-2, and thus has outperformed the performance of DCGAN, GSAFS-OQNN, MAGRU, and E-GraphSAGE techniques. Considering the precision of attack detection, the developed model achieved 92.4 %, which is better compared to graph-based and adversarial networks. The recall or sensitivity measure is close to 96.1 %, indicating better resistance to class imbalance issues. Thus, it proved that the developed model outperforms existing intrusion detection systems across various evaluation metrics, demonstrating its superior accuracy, precision, and overall effectiveness. Through the application of advanced evaluation techniques, the model's capabilities have been thoroughly validated, confirming its position as a more robust and reliable solution for intrusion detection compared to traditional systems. This proves that the proposed model is not only competitive but also offers significant improvements in the field of network security.”

Table 7 states a comparison study against well-defined reinforcement learning techniques involved in ID. RL techniques can handle large datasets and achieve less complexity. However, the developed model has resulted in higher detection performance than various RL techniques. The comparative RL techniques have not performed well under complex datasets like intrusion, and the newly developed MDDHN-AM model has achieved better performance, and thus it can be used for real-time detection. According to the Sensitivity measure, the proposed MDDHN-AM has outscored the RL techniques by 15.9 %, 13.5 %, 11.6 %, and 8.9 % over DRL, TA-NIDS, Q-DRL, and MRLC, respectively. Thus, the results proved that the hybrid architecture combining temporal convolution and recurrent neural networks helps in better understanding the sequential nature of network data, while dilated convolutions preserve essential spatial features, further enhancing the detection capabilities.

6.12. ROC-AUC estimation on the proposed framework

“Receiver Operating Characteristic (ROC) curve evaluation is given in Fig. 17.” Here, the ROC value given by the developed model is better than other models, and DBN attains the lowest score. Thus, it concluded that the developed system utilizing the MDDHN-AM, along with optimized feature extraction and fusion techniques, outperforms several

Table 7

State-of-the-art evaluation with different DRL techniques.

Dataset 1 Terms	DRL [55]	TA-NIDS [56]	Q-DRL [57]	MRLC [58]	MDDHN- AM
Accuracy	82.60469	84.7676	86.65439	89.00138	96.07429
Sensitivity	80.7363	83.05964	84.86159	87.42358	96.06746
Specificity	84.77612	86.7126	88.69223	90.75875	96.07771
Precision	86.04015	87.68248	89.5073	91.33212	92.45075
FPR	15.22388	13.2874	11.30777	9.241245	3.922288
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F1-Score	83.30389	85.30848	87.12256	89.33512	94.22441
MCC	0.653303	0.696279	0.73406	0.780719	0.912913
Dataset-2					
Accuracy	83.43685	84.33402	87.09455	88.81988	96.082
Sensitivity	81.99234	82.78061	85.78811	87.25869	96.08
Specificity	85.13514	86.16541	88.59259	90.625	96.084
Precision	86.63968	87.58435	89.60864	91.49798	96.08384
FPR	14.86486	13.83459	11.40741	9.375	3.916
FNR	18.00766	17.21939	14.21189	12.74131	3.92
NPV	80.08475	80.9322	84.46328	86.01695	96.08016
FDR	13.36032	12.41565	10.39136	8.502024	3.916157
F1-Score	84.25197	85.11475	87.65677	89.32806	96.08192
MCC	0.669256	0.68731	0.742261	0.776991	0.92164

existing ID systems. The model's robust performance in comparison to traditional methods and heuristic algorithms highlights its effectiveness in providing accurate, reliable, and efficient intrusion detection, making it a better choice for securing modern network systems.

6.13. State-of-the-art validation of different feature selection techniques

Table 8 explains a comparative study of the proposed method against some various feature selection techniques. Here, according to the F1-score, the proposed MDDHN-AM has scored 12.7 %, 10.2 %, 7.6 %, and 4.9 % higher than CCIHBO, filter-based, RNS, and TON-IOT, respectively. Most of the feature selection techniques have considered either an optimization technique or a single feature selection mechanism for selecting the intrusion-related features. However, using the single feature selection mechanism can result in reduced performance, as it cannot capture the interdependencies among the features. The developed weighted feature fusion model can process fused feature sets with well-defined characteristics and result in better detection accuracy.

6.14. Generalizability analysis on unseen datasets like NSL-KDD, UNSW-NB15, and AWID-CLS-R datasets

Table 9 represents the validation of the developed ID system against a heuristics trained under NSL-KDD, UNSW-NB-15, and AWID-CLS-R datasets, and Table 10 presents the comparative study against the deep learners under the same setup. The developed RCMPA-MDDHN-AM achieved a precision of 96.7 %, 97.3 %, and 96.6 % under NSL-KDD, UNSW-NB-15, and AWID-CLS-R datasets, showing its ability to perform well under different unseen data. This indicates the newly developed model is a promising candidate in ID research and can be implemented for real-time scenarios. When compared to other conventional models, the MDDHN-AM-based ID system has high scores. This improvement signifies the effectiveness of combining feature extraction, optimization, and hybrid deep learning models in achieving more reliable and efficient intrusion detection.

6.15. The multi-class setting of ID under NSL-KDD, UNSW-NB15, and AWID-CLS-R datasets

In this analysis, the developed RCMPA-MDDHN-AM is done under a multi-class setting for the data available in the NSL-KDD Dataset, UNSW-NB15 Dataset and AWID-CLS-R Dataset, respectively, and the results are shown as a cost function under Fig. 18. Here, four different class settings are evaluated as Normal, Flooding, Injection, and Impersonation for the NSL-KDD dataset. The UNSW-NB15b shows different attacks regulated under Normal, Neptune, Waremaster, NMPGattack, and msan,

Table 8

Evaluation of recent feature selection models.

Terms	CCIHBO [70]	Filter- based feature selection [71]	RNS [72]	TON-IOT [73]	MDDHN- AM with weighted feature fusion
Dataset-1					
Accuracy	87.37037	89.13228	90.93474	92.77513	96.07429
Sensitivity	87.36177	89.10847	90.92593	92.76786	96.06746
Specificity	87.37467	89.14418	90.93915	92.77877	96.07771
Precision	77.57738	80.40821	83.38185	86.52885	92.45075
FPR	12.62533	10.85582	9.060847	7.22123	3.922288
FNR	12.63823	10.89153	9.074074	7.232143	3.93254
NPV	93.25557	94.24276	95.24799	96.24869	97.9945
FDR	22.42262	19.59179	16.61815	13.47115	7.549247
F1-Score	82.17936	84.53507	86.99064	89.53981	94.22441
MCC	0.727585	0.764306	0.802312	0.841507	0.912913

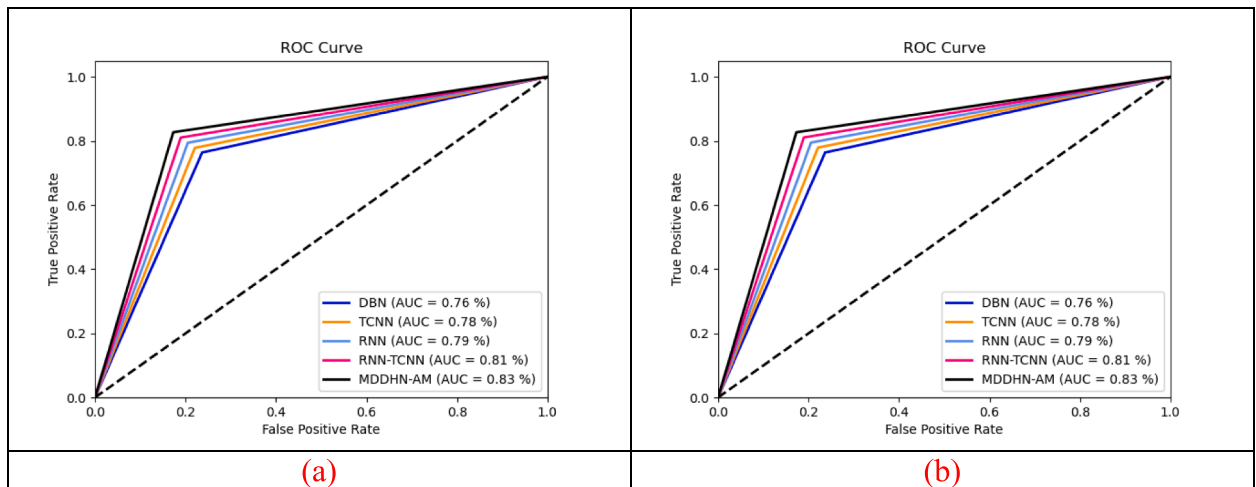
**Fig. 17.** ROC Evaluation on the Developed Model based on (a) dataset-1, and (b) Dataset-2'.

Table 9
Generalizability evaluation against recent heuristics.

Terms	OOA-MDDHN-AM [67]	TFMOA-MDDHN-AM [69]	HGSO-MDDHN-AM [68]	CMBA-MDDHN-AM [59]	RCMPA-MDDHN-AM
NSL-KDD					
Accuracy	84.02271	88.32117	90.10543	93.10624	95.94485
Sensitivity	81.51515	85.80153	87.59571	91.64038	95.18459
Specificity	86.91099	91.17647	92.93103	94.65776	96.72131
Precision	87.76509	91.68026	93.31158	94.77977	96.73736
FPR	13.08901	8.823529	7.068966	5.342237	3.278689
FNR	18.48485	14.19847	12.40429	8.359621	4.815409
NPV	80.32258	85	86.93548	91.45161	95.16129
FDR	12.23491	8.319739	6.688418	5.220228	3.262643
F1-Score	84.52474	88.64353	90.36335	93.18364	95.95469
MCC	0.682567	0.76829	0.803868	0.862648	0.919023
UNSW-NB15					
Accuracy	86.13139	88.15896	90.42985	93.99838	96.43147
Sensitivity	83.56164	85.19637	89.08228	92.42902	95.51282
Specificity	89.0625	91.5937	91.84692	95.65943	97.37274
Precision	89.70588	92.15686	91.99346	95.75163	97.38562
FPR	10.9375	8.406305	8.153078	4.340568	2.627258
FNR	16.43836	14.80363	10.91772	7.570978	4.487179
NPV	82.6087	84.219	88.88889	92.27053	95.49114
FDR	10.29412	7.843137	8.006536	4.248366	2.614379
F1-Score	86.52482	88.54003	90.51447	94.061	96.44013
MCC	0.724692	0.765827	0.809058	0.880553	0.928812
AWID-CLS-R					
Accuracy	86.5369	87.75345	91.07867	93.26845	95.86375
Sensitivity	85.90504	86.6373	90.66265	92.63158	95.586
Specificity	87.29875	89.13043	91.56415	94.01408	96.18056
Precision	89.07692	90.76923	92.61538	94.76923	96.61538
FPR	12.70125	10.86957	8.435852	5.985915	3.819444
FNR	14.09496	13.3627	9.337349	7.368421	4.414003
NPV	83.70497	84.39108	89.36535	91.5952	95.02573
FDR	10.92308	9.230769	7.384615	5.230769	3.384615
F1-Score	87.46224	88.65515	91.62861	93.68821	96.09793
MCC	0.729925	0.754634	0.821037	0.865049	0.917038

respectively. Meanwhile, the AWID-CLS-R Dataset is evaluated for three different settings here. The proposed RCMPA-MDDHN-AM model was able to detect different attack types with much higher accuracy and precision. Under the multi-class setting, the developed RCMPA-MDDHN-AM achieved an accuracy of 95.9, 96.8 %, and 95.8 % under NSL-KDD, UNSW-NB-15, and AWID-CLS-R datasets, respectively, showcasing its ability to perform under different network attack types. The comparison with other models further confirmed the effectiveness of the proposed approach, as it achieved an enhanced performance, showcasing its ability to better handle the complexities of intrusion detection tasks. By optimizing the weights through the RCMPA and performing optimal weighted feature fusion, the system effectively minimized the cost function, indicating improved efficiency and accuracy in the detection process.”.

6.16. Computational efficiency evaluation

Table 11 shows the analysis in terms of memory and time requirements. Here, as indicated, the developed RCMPA-MDDHN-AM required low memory requirements compared to existing ones, and also, the time required for the training is comparatively less. This indicates the model to achieve a faster detection rate, with fewer computational resources. This enhanced performance suggests that the proposed ID system can better detect anomalies and intrusions, offering a more reliable solution for network security. This suggests that the integration of the RCMPA optimization and the MDDHN-AM model significantly enhances the system’s ability to accurately classify network activities. Table 12 explains the computational complexity of the proposed model against some recent optimization techniques. Here, the

Table 10
Generalizability evaluation against recent deep learning models.

Terms	Dual Experience Replay [70]	SMOTE [74]	ML-SMOTE [75]	XG-Boost-SMOTE [76]	MDDHN-AM
NSL-KDD					
Accuracy	84.83374	88.64558	91.56529	93.10624	95.94485
Sensitivity	82.56881	86.32873	90.20537	90.99379	95.18459
Specificity	87.39206	91.23711	93	95.41596	96.72131
Precision	88.09135	91.68026	93.14845	95.59543	96.73736
FPR	12.60794	8.762887	7	4.584041	3.278689
FNR	17.43119	13.67127	9.794629	9.006211	4.815409
NPV	81.6129	85.64516	90	90.64516	95.16129
FDR	11.90865	8.319739	6.85155	4.404568	3.262643
F1-Score	85.24073	88.92405	91.65329	93.23787	95.95469
MCC	0.698324	0.774455	0.831769	0.863251	0.919023
UNSW-NB15					
Accuracy	85.56367	88.48337	91.07867	95.13382	96.43147
Sensitivity	83.07927	85.71429	89.84127	93.94904	95.51282
Specificity	88.38821	91.65217	92.37148	96.36364	97.37274
Precision	89.05229	92.15686	92.48366	96.40523	97.38562
FPR	11.61179	8.347826	7.628524	3.636364	2.627258
FNR	16.92073	14.28571	10.15873	6.050955	4.487179
NPV	82.1256	84.86312	89.69404	93.88084	95.49114
FDR	10.94771	7.843137	7.51634	3.594771	2.614379
F1-Score	85.96215	88.8189	91.14332	95.16129	96.44013
MCC	0.713225	0.77193	0.821952	0.902994	0.928812
AWID-CLS-R					
Accuracy	86.29359	90.59205	92.61963	95.13382	95.86375
Sensitivity	84.21053	89.61424	92.15686	94.42771	95.586
Specificity	89.0566	91.77102	93.15789	95.95782	96.18056
Precision	91.07692	92.92308	94	96.46154	96.61538
FPR	10.9434	8.22898	6.842105	4.042179	3.819444
FNR	15.78947	10.38576	7.843137	5.572289	4.414003
NPV	80.96055	87.99314	91.08062	93.65352	95.02573
FDR	8.923077	7.076923	6	3.538462	3.384615
F1-Score	87.50924	91.23867	93.06931	95.43379	96.09793
MCC	0.726497	0.811504	0.851976	0.902502	0.917038

computational efficiency is calculated based on Big O notation, indicated in terms of maximum iteration, population size, and chromosome length. Compared to the existing ones, the model has achieved better performance with low complexity. This allows the system to perform better in complex scenarios and noisy conditions. The model has the ability to handle large data traffic, with less computational requirement, making it a suitable candidate for large NID systems.”.

6.17. Sensitivity validation on the proposed framework

Sensitivity validation is provided in Fig. 19, aiding in assessing the model’s robustness. Here, this figure provides the evaluation by varying population size, such as 10, 20, 30, 40, and 50, and by varying exploration–exploitation balance (α), like 0.1, 0.2, 0.3, 0.4, and 0.5. Thus, these outcomes prove that the proposed RCMPA aids in maintaining a stable accuracy and consistent convergence speed by varying parameters. Minor fluctuations in performance were observed only under extreme parameter choices, which prove that the developed optimization algorithm is effective, reliable, and robust when selecting the optimal weights for each feature. This also confirms the method’s applicability in real-world scenarios. Table 13 shows the parameters of the RCMPA.

6.18. Convergence speed and accuracy validation on the suggested framework

A comprehensive comparison validation is carried out on the proposed framework in terms of convergence speed and detection accuracy, which helps to determine the effectiveness of the RCMPA. Here, the

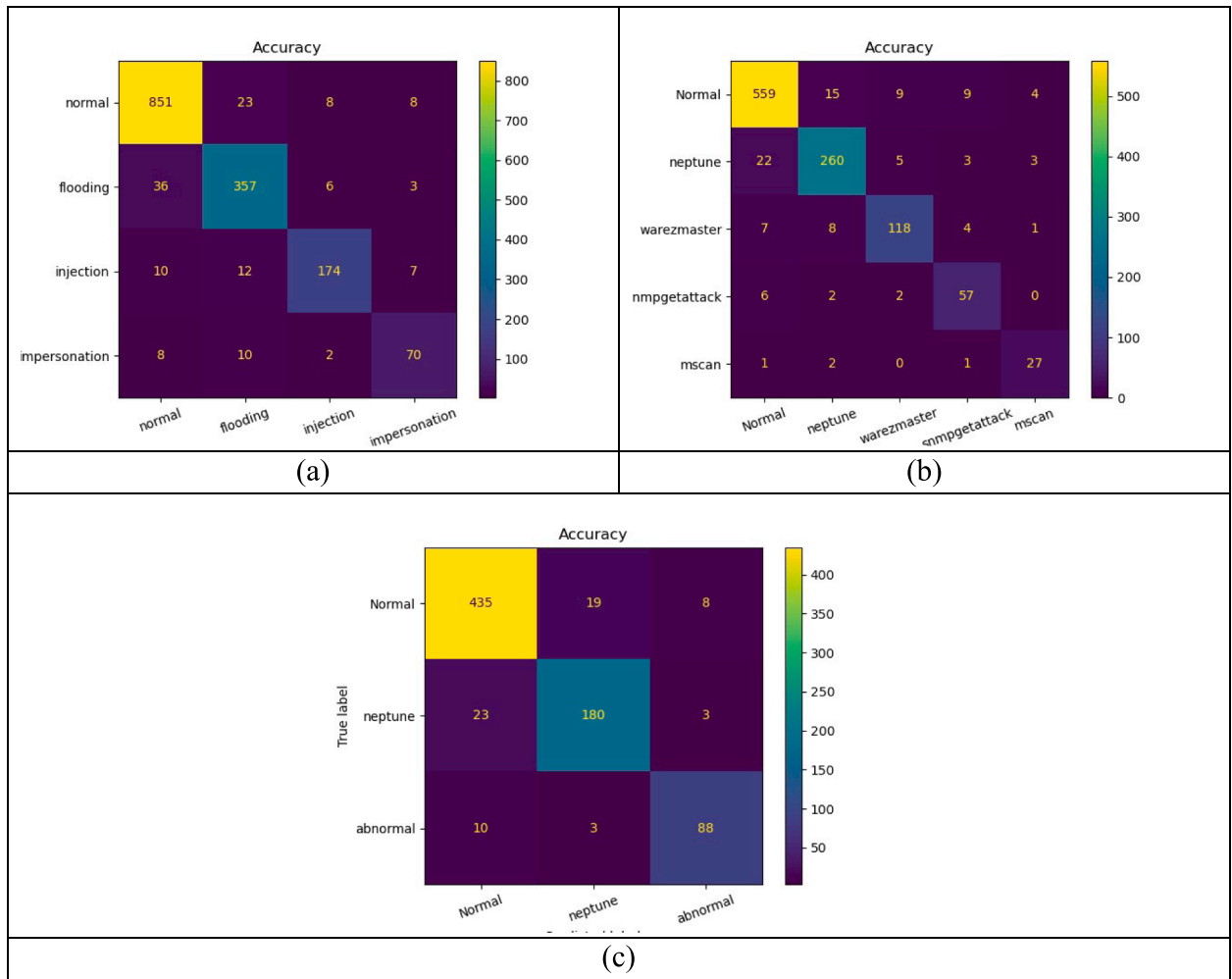


Fig. 18. Cost Function Evaluation on (a) NSL-KDD Dataset, (b) UNSW-NB15 Dataset, and (c) AWID-CLS-R Dataset”.

Table 11
Computational complexity.

Space complexity (kb)				
OOA-MDDHN-AM [67]	TFMOA-MDDHN-AM [69]	HGSO-MDDHN-AM [68]	CMBA-MDDHN-AM [59]	RCMPA-MDDHN-AM
43	39	37	32	31
DBN [33]	TCNN [63]	RNN [64]	RNN-TCNN [63 64]	MDDHN-AM
46	43	54	32	31
Time Complexity (mins)				
OOA-MDDHN-AM [67]	TFMOA-MDDHN-AM [69]	HGSO-MDDHN-AM [68]	CMBA-MDDHN-AM [59]	RCMPA-MDDHN-AM
31.4	29.7	30.4	28.9	27.05
DBN [20]	TCNN [38]	RNN [39]	RNN-TCNN [38 39]	MDDHN-AM
31.4	32.3	30.4	28.98	27.05

existing optimization algorithms, such as GWO, WOA, and FA, are used for the comparison. Based on Table 14, the detection accuracy of the developed RCMPA is 95 % and the existing optimizers, such as GWO, WOA, and FA, give the accuracy to be 93 %, 94 %, and 90 %, respectively. When considering the convergence speed, the RCMPA attained 8/10, where the scores of existing GWO, WOA, and FA out of 10 are 5, 7,

Table 12
Computational complexity.

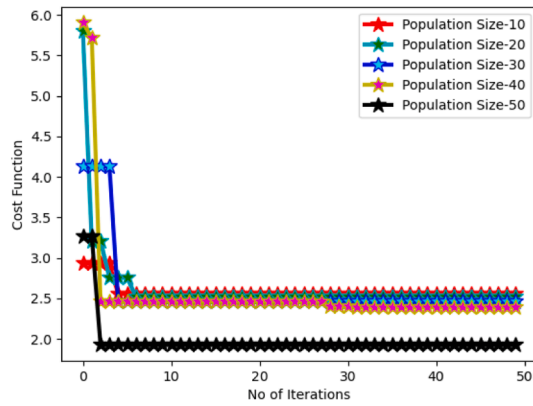
Technique	Computational efficiency
OOA-MDDHN-AM [67]	$O[MaxIt + 2 + Pp + 3 + Lch + 3]$
TFMOA-MDDHN-AM [69]	$O[MaxIt + 3 + Pp + 1 + Lch + 4]$
HGSO-MDDHN-AM [68]	$O[MaxIt + 2 + Pp + 4 + Lch + 2]$
CMBA-MDDHN-AM [59]	$O[MaxIt + 2 + Pp + 2 + Lch + 1]$
RCMPA-MDDHN-AM	$O[MaxIt + 1 + Pp + Lch + 1]$

and 6. Thus, these results clearly show the enhanced performance of the developed RCMPA in attaining an optimal solution by using its faster converging speed. This also validates its practical applicability, effectiveness, and robustness.

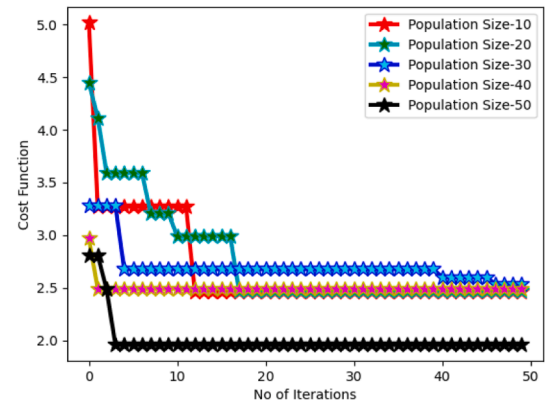
7. Conclusion

The newly developed ID model was designed to predict abnormal activities within networks to enhance security. It presented an innovative model, named MDDHN-AM, for intrusion detection infused with the RCMPA for weight optimization. By leveraging the power of deep features and statistical features, the developed MDDHN-AM accurately captured the temporal and spatial features in the network traffic by tackling the data imbalance issue. The developed RCMPA strategy was employed to optimize the weights to maximize the correlation coefficient. The prediction was executed using the MDDHN-AM method and finally provided the detected outcome. The RCMPA-MDDHN-AM-based

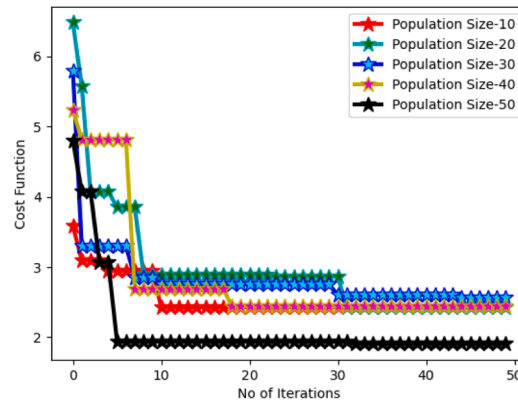
Sensitivity Evaluation by Varying Population Size



Dataset-1

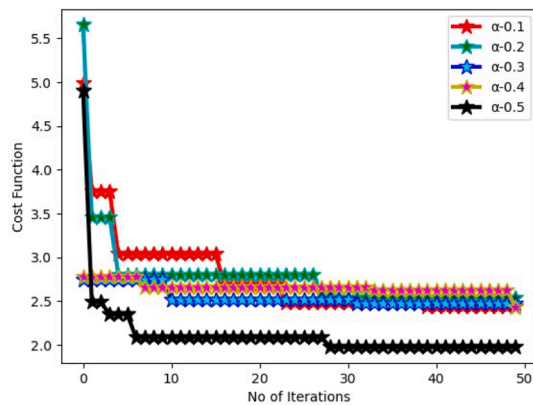


Dataset-2

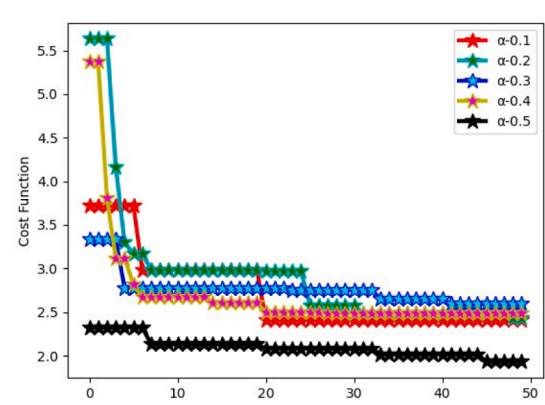


Dataset-3

Sensitivity Evaluation by Varying Exploration–Exploitation Balance



Dataset-1



Dataset-2

Fig. 19. Sensitivity evaluation on the developed model.

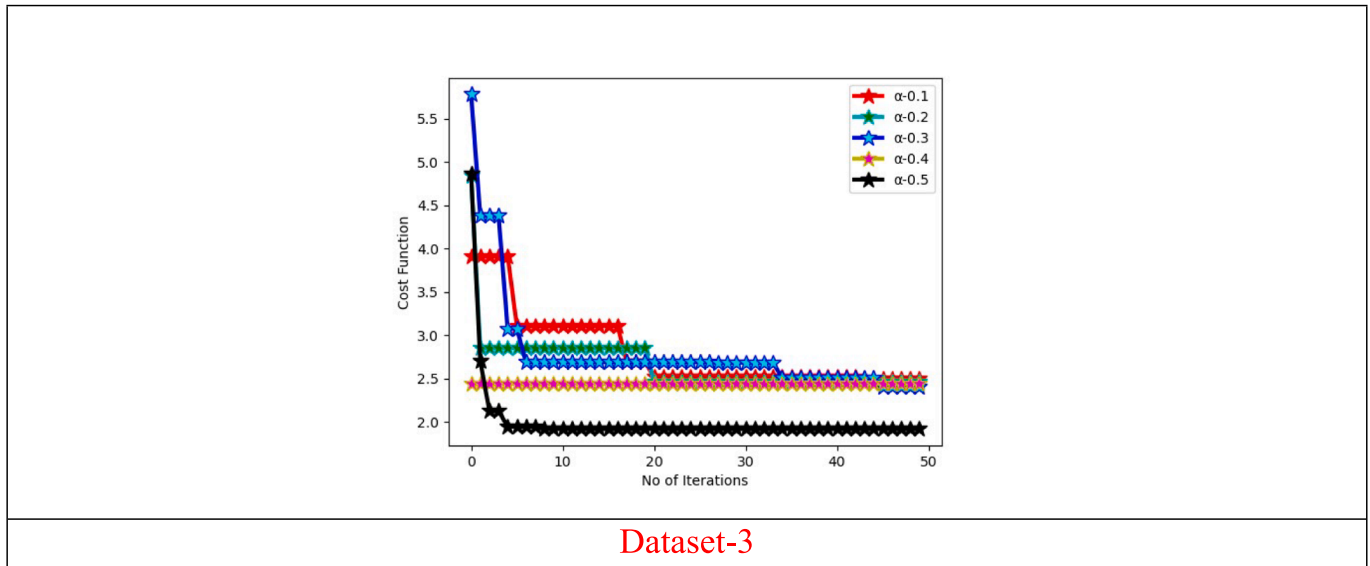


Fig. 19. (continued).

Table 13
Algorithmic parameters.

Parameter	Description	Typical value/ range
Population Size (N)	Total candidate solution in the lookup region.	10
Number of Iterations (MaxIter)	Maximum iteration	100
Mask Efficiency Factor (MEF)	It indicates the required weight of MEF when calculating the fitness function	0.3
Protection Coverage Factor (PCF)	It is the mask utilization's and spatial coverage's weight factor	0.2
Infection Risk Factor (IRF)	Required weight for tuning the infection risk possibility	0.2
Exploration–Exploitation Balance (α)	Indicating the trade-off present between the local exploitative and global explorative phases	0.3
Randomization Factor (β)	It is the stochastic perturbation's degree, aids in avoiding the local optima	0.1
Crossover / Combination Rate	Probability of combining solutions in the population	0.6
Mutation / Perturbation Rate	Probability of random variation applied to solutions	0.1

Table 14
Convergence speed and accuracy evaluation on the developed model.

Algorithm	Convergence speed (1–10)	Detection accuracy (%)
GWO [46]	7	93
WOA [47]	5	94
FA [48]	6	90
RCMPA	8	95

ID framework specified greater accuracy of 5.79 % than DBN, 4.05 % than TCNN, 8.57 % than RNN, and 10.59 % than TCNN-RNN using dataset-2. Thus, it concluded that the proposed RCMPA-MDDHN-AM-based ID system demonstrated superior performance in detecting network intrusions compared to existing models such as DBN, TCNN, RNN, and TCNN-RNN. By effectively combining the power of TCNN, RNN, and an AM, the MDDHN-AM model significantly enhanced the detection accuracy and robustness. The integration of the RCMPA

optimization method further improved the feature fusion process, maximizing the correlation between the features and contributing to the model's overall efficacy. The results highlight the potential of the developed system to address challenges in network security, providing a more accurate and efficient solution for detecting abnormal network activities. The success of the model suggested its applicability in real-world cybersecurity applications, where quick and precise intrusion detection is critical.

Applicability of Findings: The developed model has the ability to handle imbalanced, high-dimensional, along with sequential system traffic as it uses the power of MDDHN-AM and RCMPA. It aids in enhancing the accuracy of intrusion detection, while reducing the false scores, thus making the model more adaptable to IoT, cloud contexts, as well as enterprise networks. It also allows for real-time surveillance, intrusion detection in terms of enhanced scalability, and automatic prioritization of the given feature, thus minimizing the computational overhead. Furthermore, the proposed system can offer a systematically blueprint to the researchers, making the model adaptable to varying network circumstances, thus proving its theoretical and practical relevance.

Limitations and Future Scope: Despite the significant performance improvements, the model's complexity can be a limitation. The hybridization of TCNN and RNN increases computational overhead, which may lead to longer training times and increased resource consumption. Additionally, the dependency on a large, diverse dataset for training can also affect the model's generalization ability if sufficient data is not available or if the data is biased. So, future work can focus on further optimizing the model to reduce computational complexity without compromising accuracy. Techniques such as model pruning, quantization, or the use of lightweight neural architectures can help address these challenges. Moreover, extending the model to handle a wider variety of network environments and attack types will be crucial for its deployment in real-world systems. Also, in the future, the image type data and its applicability in NID will be evaluated to make the developed model suitable for a wider range of data types. Finally, incorporating a pre-processing process could also allow the model to be trained across decentralized data sources, preserving privacy while enhancing its robustness."

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Ayantayo Abiodun, Kaur Amrit, Kour Anit, Schmoor Xavier, Shah Fayyaz, Vickers Ian, Kearney Paul, Abdelsamea Mohammed M. Network intrusion detection using feature fusion with deep learning. *J Big Data* 2023;10(1).
- [2] Logeswari G, Bose S, Anitha TJIA. An intrusion detection system for SDN using machine learning. *Intell Automat Soft Comput* 2023;35(1):867–80.
- [3] Sajid Muhammad, Malik Kaleem Razzaq, Almogren Ahmad, Malik Tauqeer Safdar, Khan Ali Haider, Tanveer Jawad, Rehman Ateeq Ur. Enhancing intrusion detection: a hybrid machine and deep learning approach. *J Cloud Comput* 2024;13(1).
- [4] Kasongo SM. A deep learning technique for intrusion detection system using a recurrent neural networks based framework. *Comput Commun* 2023;199:113–25.
- [5] Siddiqi MA, Pak W. Tier-based optimization for synthesized network ID system. *IEEE Access* 2022;10:108530–44.
- [6] Pontes CFT, de Souza MMC, Gondim JJC, Bishop M, Marotta MA. A New method for flow-based network ID using the inverse pots model. *IEEE Trans Netw Serv Manag* June 2021;18(2):1125–36.
- [7] Musleh Dhiaa, Alotaibi Meera, Alhaidari Fahd, Rahman Atta, Mohammad Rami M. Intrusion detection system using feature extraction with machine learning algorithms in IoT. *J Sens Actuat Networks* 2023;12(2).
- [8] Maayah B, Arqub OA. Uncertain M-fractional differential problems: existence, uniqueness, and approximations using Hilbert reproducing technique provisioner with the case application: series resistor-inductor circuit. *Phys Scr Feb.* 2024;99(2): 025220.
- [9] Arqub OA, Mezghiche R, Maayah B. Fuzzy M-fractional integrodifferential models: theoretical existence and uniqueness results, and approximate solutions utilizing the Hilbert reproducing kernel algorithm. *Front Phys* 2023;11:1252919.
- [10] Arqub OA. Adaptation of reproducing kernel algorithm for solving fuzzy Fredholm–Volterra integrodifferential equations. *Neural Comput Appl Jul.* 2017; 28(7):1591–610.
- [11] Arqub OA, Abukhaled M, Sweis H, Shawagfeh N. Numerical analysis of Lucas polynomials innovative technique within the Galerkin strategy for solving fuzzy-type differential models with an application in the electrical circuit engineering field. *Results Eng* 2025.
- [12] Xiao M, Zhou Q, Zhang Z, Yin J. Real-time intrusion detection in power grids using deep learning: ensuring dpu data security. *HighTech Innov J* 2024;5(3):814–27.
- [13] Farda I, Thammano A. An adaptive differential evolution with multiple crossover strategies for optimization problems. *HighTech Innovat J* 2024;5(2):231–58.
- [14] Ouiazzane O, Malika Addou S, Barramou F. A Suricata and machine learning based hybrid network intrusion detection system. *International Conference on Information, Communication & Cybersecurity*, 2021, pp. 474–485.
- [15] Pavithra S, Durgadevi P. An approach for network based intrusion detection system using Snort. *Data Sci Intell Comput Tech* 2021;481–49.
- [16] Jayaraj R, Pushpalatha A, Sangeetha K, Kamaleshwar T, Udhaya Shree S, Damodaran Deepa. Intrusion detection based on phishing detection with machine learning. *Measur: Sens* 2024;31.
- [17] Qazi Emad Ul Haq, Faheem Muhammad Hamza, Zia Tanveer. HDLNIDS: hybrid deep-learning-based network intrusion detection system. *Appl Sci* 2023;13(8).
- [18] Sharma B, Sharma L, Lal C, Roy S. Anomaly based network intrusion detection for IoT attacks using deep learning technique. *Comput Electr Eng* 2023;107.
- [19] Lopez-Martin M, Sanchez-Esguevilas A, Arribas JI, Carro B. Network ID based on extended RBF neural network with offline reinforcement learning. *IEEE Access* 2021;9:153153–70.
- [20] Srilatha D, Thillaiarasu N. Implementation of Intrusion detection and prevention with deep learning in cloud computing. *J Inform Technol Manag* 2023;15(Special Issue):1–18.
- [21] Bowen B, Chennamaneni A, Goulart A, Lin D. BLoCNet: a hybrid, dataset-independent intrusion detection system using deep learning. *Int J Inf Secur* 2023; 22(4):893–917.
- [22] Sarikaya A, Günel Kılıç B, Demirci M. Raids: Robust autoencoder-based intrusion detection system model against adversarial attacks. *Comput Secur* 2023;135: 103483.
- [23] Wang C, et al. Intrusion detection system based on one-class support vector machine and Gaussian mixture model. *Electronics* 2023;12(4):930.
- [24] Van Langendonck L. PPT-GNN: A practical pre-trained temporal graph neural network for intrusion detection. M.S. thesis. Universitat Politècnica de Catalunya; 2024.
- [25] Altunay HC, Albayrak Z. A hybrid CNN+ LSTM-based intrusion detection system for industrial IoT networks. *Eng Sci Technol Int J* 2023;38:101322.
- [26] Khan NW, et al. A hybrid deep learning-based intrusion detection system for IoT networks. *Math Biosci Eng* 2023;20(8):13491–520.
- [27] Hnamte Vanlalruata, Hussain Jamal. DCNNBiLSTM: An efficient hybrid deep learning-based intrusion detection system. *Telematics and Informatics Reports*. 10, 2023.
- [28] Omer N, Samak AH, Taloba AI, Rasha Abd El-Aziz M. A novel optimized probabilistic neural network approach for intrusion detection and categorization. *Alex Eng J* 2023;72:351–61.
- [29] Siliveri Arun Kumar, Rao Kovvur Ram Mohan, Solleti Ramana, Suresh Kumar LK, Madhu Bhukya. A model for multi-attack classification to improve intrusion detection performance using deep learning approaches. *Measurement: Sensors*. 30, 2023.
- [30] Xie J, Wang H, Garibaldi JM, Wu D. Network ID based on dynamic intuitionistic fuzzy sets. *IEEE Trans Fuzzy Syst Sept.* 2022;30(9):3460–72.
- [31] An X, Lü X, Yang L, Zhou X, Lin F. Node state monitoring scheme in fog radio access networks for ID. *IEEE Access* 2019;7:21879–88.
- [32] Zhong W, Yu N, Ai C. Applying big data based deep learning system to ID. *IEEE Big Data Mining and Analytics September* 2020;3(3):181–95.
- [33] Wang Z, Zeng Y, Liu Y, Li D. Deep belief network integrating improved kernel-based extreme learning machine for network ID. *IEEE Access* 2021;9:16062–91.
- [34] Khan FA, Gumaei A, Derhab A, Hussain A. A novel two-stage deep learning model for efficient network ID. *IEEE Access* 2019;7:30373–85.
- [35] Ashiku Lirim, Dagli Cihan. Network ID system using deep learning. *Procedia Comput Sci* 2021;185:239–47.
- [36] Ahmed U, Lin J-W, Srivastava G. A resource allocation deep active learning based on load balancer for network ID in SDN sensors. *Comput Commun* 2022;184: 56–63.
- [37] Zhihao Wang, Jiang Dingde, Huo Liuwei, Yang Wei. An efficient network ID approach based on deep learning. *Wireless Networks*. 9(545), 2021.
- [38] Wang X, Yin S, Li H, Wang J, Teng L. A network ID method based on deep multi-scale convolutional neural network. *Int J Wireless Inf Networks* 2020;27:503–17.
- [39] Maddu M, Rao YN. Network intrusion detection and mitigation in SDN using deep learning models. *Int J Inf Secur* 2024;23(2):849–62.
- [40] Aljehane, Nojood O, Mengash Hanan A, Hassine Siwar BH, Alotaibi Faiz A, Salama Ahmed S, Abdelbagi Sitebanat. Optimizing intrusion detection using intelligent feature selection with machine learning model. *Alexand Eng J* 2024;91.
- [41] Aljehane NO, Mengash HA, Eltahir MM, Alotaibi FA, Aljameel SS, Yafoz Ayman, Alsini Raed, Assiri Mohammed. Golden jackal optimization algorithm with deep learning assisted intrusion detection system for network security. *Alex Eng J* 2024; 86:415–24.
- [42] Saleh HM, Marouane H, Fakhfakh A. Stochastic gradient descent intrusions detection for wireless sensor network attack detection system using machine learning. *IEEE Access* 2024.
- [43] Karthikeyan M, Manimegalai D, RajaGopal K. Firefly algorithm based WSN-IoT security enhancement with machine learning for intrusion detection. *Sci Rep* 2024; 14.
- [44] Ullah S, Boulila W, Koubaa A, Ahmad Jawad. MAGRU-IDS: A multi-head attention-based gated recurrent unit for intrusion detection in IIoT networks. *IEEE Access* 2023.
- [45] Mirlashari M, Rizvi SAM. Enhancing IoT intrusion detection system with modified E-GraphSAGE: a graph neural network approach. *Int J Inf Technol* 2024;16(4): 2705–13.
- [46] Khatami, Sarvenaz Sadat, Mehrdad Shoeibi, Anita Ershadi Oskoue, Diego Martin, and Maral Keramat Dashlboroun. SDGWO-GAN: A novel five-dimensional gray wolf optimizer for generative adversarial network-enabled intrusion detection in IoT systems. *Comput Mater Continua* 2025;82(1).
- [47] Abualhaj Moleh M, Al-Khatib Sumaya N, Al Zyouud Mahran, Qaddara Iyas, Anbar Mohammed. Enhancing intrusion detection system performance using a hybrid of harris hawks and whale optimization algorithms. *Engineering Technol Appl Sci Res* 2025;15(4):24354–24361.
- [48] Ogunseyi Taiwo Blessing, Thyagarajan Gogulakrishnan. An explainable LSTM-based intrusion detection system optimized by firefly algorithm for IoT networks. *Sensors* 2025;25(7).
- [49] Zhou W, Xia C, Wang T, Liang X, Lin W, Li X, et al. Hidim: a novel framework of network intrusion detection for hierarchical dependency and class imbalance. *Comput Secur* 2025;148:104155.
- [50] Liu T, Fu Y, Wang K, Duan X, Wu Q. A multiscale approach for network intrusion detection based on variance-covariance subspace distance and EQL v2. *Comput Secur* 2025;148:104173.
- [51] Cai S, Zhao Y, Lyu J, Wang S, Hu Y, Cheng M, et al. DDP-DAR: Network intrusion detection based on denoising diffusion probabilistic model and dual-attention residual network. *Neural Netw* 2025;184:107064.
- [52] Abdulganiyu OH, Tchakouch TA, Saheed YK, Ahmed HA. XIDINTFL-VAE: XGBoost-based intrusion detection of imbalance network traffic via class-wise focal loss variational autoencoder. *J Supercomput* 2025;81(1):1–38.
- [53] Thanathamath P, Sawangreerak S, Chantamunee S, Nizam DNM. SHAP-instance weighted and anchor explainable AI: enhancing XGBoost for financial fraud detection. *Emerg Sci J* 2024;8:2404–30.
- [54] Mahmud Bahar Uddin, Al Mamun Abdullah, Hossen Md Jakir, Yue Hong Guan, Jahan Busrat. Light-weight deep learning model for accelerating the classification of mango-leaf disease. *Emerg. Sci. J.* 2024;8(1):28–42.
- [55] Sharma A, Singh M. Batch reinforcement learning approach using recursive feature elimination for network intrusion detection. *Eng Appl Artif Intel* 2024;136:109013.
- [56] He M, Wang X, Wei P, Yang L, Teng Y, Lyu R. Reinforcement learning meets network intrusion detection: a transferable and adaptable framework for anomaly behavior identification. *IEEE Trans Netw Serv Manag* 2024;21(2):2477–92.
- [57] Mesadieu F, Torre D, Chennamaneni A. Leveraging deep reinforcement learning technique for intrusion detection in SCADA infrastructure. *IEEE Access* 2024.
- [58] Darabi B, Bag-Mohammadi M, Karami M. A micro reinforcement learning architecture for intrusion detection systems. *Pattern Recogn Lett* 2024;185:81–6.
- [59] Yuan Y, Shen Q, Wang S, Ren J, Yang D, Yang Q, et al. Coronavirus mask protection algorithm: a new bio-inspired optimization algorithm and its applications. *J Bionic Eng* 2023;20:1747–65.
- [60] Chen S, Jianbo Yu, Wang S. One-dimensional convolutional neural network-based active feature extraction for fault detection and diagnosis of industrial processes and its understanding via visualization. *ISA Trans March* 2022;122:424–43.

- [61] Vasanthakumar GU, Adnan MM, Soujanya T, Hafidh I, Sadi Ai. Network traffic anomaly detection using t-distributed stochastic neighbor embedding and variational autoencoder approach. *IEEE* 2024;1–4.
- [62] Fan L, Shen H, Xie F, Su J, Yu Y, Hu D. DC-tCNN: a deep model for EEG-based detection of dim targets. *IEEE Trans Neural Syst Rehabil Eng* 2022;30:1727–36.
- [63] Belhadi A, Djenouri Y, Djenouri D, Srivastava G, Lin J-W. Group intrusion detection in the internet of things using a hybrid recurrent neural network. *Clust Comput* 2023;26(2):1147–58.
- [64] Farabi A, Shad MR, Khandaker I. IntrusionX: A hybrid convolutional-LSTM deep learning framework with squirrel search optimization for network intrusion detection. *arXiv preprint arXiv:2510.00572*, 2025.
- [65] Alazab M, et al. An effective networks intrusion detection approach based on hybrid Harris Hawks and multi-layer perceptron. *Egypt Informat J* 2024;25: 100423.
- [66] Peiyuan Sun, Qiben Yan, Haoyi Zhou, Jianxin Li. Osprey: A fast and accurate patch presence test framework for binaries. *Comput Commun* 2021;173:95–106.
- [67] Zhou Pengzhen, Zhang Huifu, Liang Wei. Research on hybrid intrusion detection based on improved Harris Hawk optimization algorithm. *Connect Sci* 2003;35(1).
- [68] Panteleev A, Kolessa A. Application of the tomitt flock metaheuristic optimization algorithm to the optimal discrete time deterministic dynamical control problem. *Algorithms* 2022;15:301.
- [69] Ye Z, Luo J, Zhou W, Wang M, He Q. An ensemble framework with improved hybrid breeding optimization-based feature selection for intrusion detection. *Futur Gener Comput Syst* 2024;151:124–36.
- [70] Zouhri H, Idri A, Ratnani A. Evaluating the impact of filter-based feature selection in intrusion detection systems. *Int J Inf Secur* 2024;23(2):759–85.
- [71] Saheed YK, Kehinde TO, Ayobami Raji M, Baba UA. Feature selection in intrusion detection systems: a new hybrid fusion of Bat algorithm and Residue number system. *J Inform Telecommun* 2024;8(2):189–207.
- [72] Li J, Othman MS, Chen H, Yusuf LM. Optimizing IoT intrusion detection system: feature selection versus feature extraction in machine learning. *J Big Data* 2024;11 (1):36.
- [73] Tan H, Wang L, Zhu D, Deng J. Intrusion detection based on adaptive sample distribution dual-experience replay reinforcement learning. *Mathematics* 2024;12 (7):948.
- [74] Sayegh HR, Dong W, Al-madani AM. Enhanced intrusion detection with LSTM-Based model, feature selection, and SMOTE for imbalanced data. *Appl Sci* 2024;14 (2):479.
- [75] Widodo AO, Setiawan B, Indraswari R. Machine learning-based intrusion detection on multi-class imbalanced dataset using SMOTE. *Procedia Comput Sci* 2024;234: 578–83.
- [76] Fan Z, Sohail S, Sabrina F, Gu X. Sampling-based machine learning models for intrusion detection in imbalanced dataset. *Electronics* 2024;13(10):1878.



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